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TurboACSourceDefTopic



About the TurboACSource

About the TurboACSource

The TurboACSource instrument provides AC waveform source capabilities for testing devices such as precision A/D and D/A converters, high speed A/D and D/A converters and high performance SOC's such as set top box devices.

Typical tests performed using the TurboAC instruments include gain accuracy, frequency response, distortion, and signal-to-noise ratio.

The TurboACSource instrument can generate high-accuracy, low-noise and distortion signals with frequency components of up to 150MHz.

This section includes the following topics:

- [Features](#)
- [Safety Information](#)
- [Key Terms](#)
- [TurboAC Source Software Licensing Requirements](#)

TurboACSource information includes the following sections:

- [TurboACSource Theory of Operation](#)
- [TurboACSource Programming](#)
- [TurboACSource Debug Display](#)
- [TurboACSource Examples](#)

Related Topics

- [TurboAC Specifications](#)
- [TurboACSource \(VBT syntax\)](#)
- [TurboAC DIB Design Guidelines](#)

The TurboACSource is a dual source per each TurboAC instrument board. Each board has two TurboACCapture instruments as well. For more information, see [About the TurboACCapture](#).

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TurboACSource Examples

TurboACSource Examples

This section contains a program example for the TurboACSource instrument. This IG-XL workbook example shows all IG-XL data, VBT code, necessary for a complete running program.
The follow topic is available:

- [TurboACSource CW Plus AWG Example](#)

For more information, see:

- [TurboACSource \(VBT Syntax Reference\)](#)
- [TurboACSource Programming](#)

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[TurboACSource Examples](#) : TurboACSource CW Plus AWG Example

TurboACSource CW Plus AWG Example

This workbook example includes the following program components:

- | | |
|---|---------------------------------|
| • | DUT Connections |
| • | DataTool Sheets |
| • | VBT Code |

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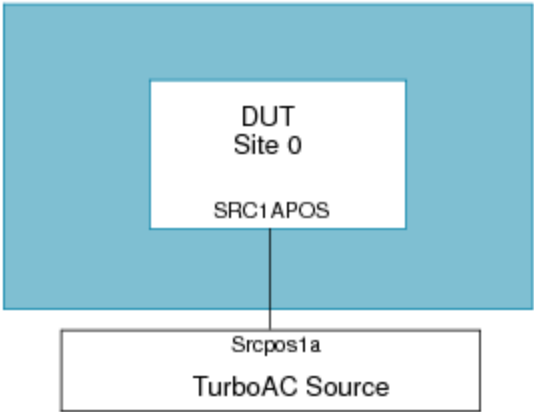
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TurboACSource Examples : TurboACSource CW Plus AWG Example : DUT Connections

DUT Connections



TurboAC Source Connections

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TurboACSource Examples : TurboACSource CW Plus AWG Example : DataTool Sheets

DataTool Sheets

This workbook example uses the following DataTool sheets:

- Pin Map Sheet
- Channel Map Sheet
- Test Instances Sheet
- Flow Table Sheet
- Mixed Signal Timing Sheet
- Wave Definitions Sheet

Pin Map Sheet

Pin Map		
Group Name	Pin Name	Type
	Src1APos	Analog

Channel Map Sheet

Channel Map

DIB ID:

View Mode: Signal

Device Under Test		Tester Channel		
Pin Name	Package Pin	Type	Site 0	Comments
Src1APos		TurboACSrcPos	11.srcpos1a	

Test Instances Sheet

Test Instances

Test Name	Test Procedure			DC Sp
	Type	Name	Called As	Category
SigCombCWPlusAWG_VBT_I	VBT	SigCombCWPlusAWGtoCaptureLoopback_VBT		

The test instance named SigCombCWPlusAWG_VBT_I has the following test-specific parameters and values as shown in the Instance Editor:

Instance Editor [VBT Test 'SigCombCWPlusAWGtoCaptureLoopback_VBT']

...

Lock

Down

Up

✓

✗

Instance: SigCombCWPlusAWG_VBT_I

Parameters

Timing and Levels

Limits

	Name	Value	
☐	SrcPin	Src1APos	...
	SrcAmp	0.5	...
	SrcFreq1	11000000	...
	SrcFreq2	9000000	...
	SrcFilt	12100000	...
	SrcWaveDef	aSine	

Flow Table Sheet

Flow Table

					Flow Domain:		
Label	Enable	Gate			Command		TName
		Job	Part	Env	Opcode	Parameter	
					Test	SigCombCWPlusAWG_VBT_I	
					Use-Limit	SigCombCWPlusAWG_V	IMD3A
					Use-Limit	SigCombCWPlusAWG_V	IMD3B

Mixed Signal Timing Sheet

Mixed Signal Timing



Set Name	Subset	Resource		Clocking					Instrument	Waveform		MSW
		Type	ID	Fs	N	Fr	M	USR	Data	Definition	Filter	Settings
Audio1KhzLoopback	Subset1	BBACSource	src1	20000	20	1000	1	1		aSine		loc=0,mnrelative
Audio1KhzLoopback	Subset1	BBACCapture	cap1	59904.936	32768	1000	547	1				

Wave Definitions Sheet

Wave Definitions



WaveDefName	WaveDefType	WaveDef Component	Repeat Count	Relative Period	Relative Amplitude	Relative Offset	Primitive Parameters
aSine	Primitive	Sine	1	1.E+00	1.E+00	0	phase=0.0

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[TurboACSource Examples](#) : [TurboACSource CW Plus AWG Example](#) : VBT Code

VBT Code

Public Function SigCombCWPlusAWGtoCaptureLoopback_VBT(SrcPin As PinList, SrcAmp As Double, SrcFreq1 As Double, SrcFreq2 As Double, _
 SrcFilt As Double, SrcWaveDef As String, _
 Optional TNames_ As String = "IMD3A|IMD3B") As Long

```
Dim IMD3A As New SiteDouble, IMD3B As New SiteDouble
Dim Bin1 As New SiteDouble, Bin2 As New SiteDouble
Dim site As Variant
Dim RMS_reading As New SiteDouble, Vpk_reading As New SiteDouble, _
    CalFactor As New SiteDouble
```

```
' Connect Turbo AC Source
TheHdw.TurboACSource(SrcPin).Connect tlTurboACSourceFromPOS Or _
    tlTurboACSourceFromNEG, tlTurboACSourceToLoopBack

' Reset the Turbo AC CW to 1MHz frequency to avoid changing filter values
' while CW is still set to higher frequency from previous test
TheHdw.TurboACSource(SrcPin).CW.Frequency = 1000000#
TheHdw.TurboACSource(SrcPin).CW.Amplitude.Peak = _
    TheHdw.TurboACSource(SrcPin).CW.Amplitude.Peak.Min

' Set up the Turbo AC Source Signal for the AWG tone at 9MHz
With TheHdw.TurboACSource(SrcPin).AWG
    .Signals.Add "SrcSig"
```

```

' set signal source to AWG + CW because this signal will be used with
' signal combination
.Signals("SrcSig").SignalSource = tlTurboACSourceSignalSourceAWGAndStdCW
.Signals("SrcSig").WaveDefinitionName = SrcWaveDef
.Signals("SrcSig").SampleRate = SrcFreq2 * 100
.Signals("SrcSig").SampleSize = 100
.Signals("SrcSig").CycleCount = 1
.Signals("SrcSig").Amplitude = SrcAmp
.Signals("SrcSig").Filter.LowPass = SrcFilt
.Signals("SrcSig").LoadSettings
End With

' Start the AWG Source
TheHdw.TurboACSource.Pins(SrcPin).AWG.Signals("SrcSig").Start tlStartImmediately

' Use RMS to DC converter to determine the above cal factor to use in the
' signal combination circuit
' At this point AWG is driving a 9MHz single tone through power combiner.
' Measure the amplitude using the RMS to DC converter readback
' RMS to DC reading is physically made into a 100 ohm differential
' termination
RMS_reading = TheHdw.TurboACSource(SrcPin).ReadOutputRMS.Math.Multiply(2) ' To
account for 100 ohm term in rms 2 dc
Vpk_reading = RMS_reading.Multiply(Sqr(2))
CalFactor = Vpk_reading.Invert.Multiply(SrcAmp)

' Stop the AWG
TheHdw.TurboACSource.Pins(SrcPin).AWG.Signals.Stop tlStopAtEnd

' Reprogram the AWG signal using the cal factor
With TheHdw.TurboACSource(SrcPin).AWG.Signals("SrcSig")
' set the signal source to AWG + CW because this signal will be used
' with signal combination
.SignalSource = tlTurboACSourceSignalSourceAWGAndStdCW
.WaveDefinitionName = SrcWaveDef
.SampleRate = SrcFreq2 * 100
.SampleSize = 100
.CycleCount = 1
For Each site In TheExec.Sites.Active

```

```

        .Amplitude = SrcAmp * CalFactor
    Next site
    .LoadSettings
End With
' Run CW through the sig combination circuit alone and cal its amplitude
With TheHdw.TurboACSource(SrcPin).CW
    .Signals.Add "CWSignal"
    ' Set signal source to AWG + CW because this signal will be used with
    ' signal combination
    .Signals("CWSignal").SignalSource = tITurboACSourceSignalSourceAWGAndStdCW
    .Signals("CWSignal").Function.Sine.Amplitude = SrcAmp
    .Signals("CWSignal").Function.Sine.Frequency = SrcFreq1
    .Signals("CWSignal").LoadImpedance = 100
    .Signals("CWSignal").Filter.LowPass = SrcFilt

    .Signals("CWSignal").LoadSettings
End With

RMS_reading = TheHdw.TurboACSource(SrcPin).ReadOutputRMS
Vpk_reading = RMS_reading.Multiply(Sqr(2))
CalFactor = Vpk_reading.Invert.Multiply(SrcAmp)

' Reprogram the CW signal using the cal factor
With TheHdw.TurboACSource(SrcPin).CW.Signals("CWSignal")
    ' Set signal source to AWG + CW because this signal will be used with
    ' signal combination
    .SignalSource = tITurboACSourceSignalSourceAWGAndStdCW
    For Each site In TheExec.Sites.Active
        .Function.Sine.Amplitude = SrcAmp * CalFactor
    Next site
    .Function.Sine.Frequency = SrcFreq1
    .LoadImpedance = 1000000# ' Set impedance to 1Mohm because measuring
        ' using Turbo capture, which is Hi-Z
    .Filter.LowPass = SrcFilt
    .LoadSettings
End With

' Start the AWG Source

```

End Function

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About the TurboACSource : Features

Features

Each TurboACSource instrument board has two TurboACSource instruments and two TurboACCapture instruments. The following is a list of basic TurboACSource features:

- Independent arbitrary waveform generator (AWG)
- 15MHz AWG bandwidth with 1Gsps max sampling rate
- 4M samples source memory
- Sample rate of 5kHz to 1GHz
- 15MHz low-pass selectable filter
- Pattern Microcode control for the AWG
- Parameter sets (PSets) programming for the AWG
- Independent continuous waveform (CW) source
- 150MHz CW bandwidth programmable to 310MHz
- Optional continuous waveform (CW) source input for IEEE signal distribution

- [Eight single-ended outputs or four differential outputs](#)
- [Single-ended and differential performance](#)
- [Single combiner for intermodulation testing](#)
- [DIB input with access to filters, combiner](#)
- [DIB access lines](#)
- [Internal analog loopback](#)
- [Capture cancellation](#)
- [Connections to CC Common](#)
- [Connections to device ground sense \(DGS\)](#)

For more information, see [TurboAC Specifications](#).

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About the TurboACSource : Safety Information

Safety Information

When operating the TurboACSource instrument, always follow general UltraFLEX test system safety guidelines. For more information, see [Test System Safety](#).

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About the TurboACSource : Key Terms

Key Terms

AUX input	The port used by either the Optional CW or Optional LO.
BBAC	Broadband Analog Channel
CCC	Channel Card Common
compatibility port	Port 1 of each TurboAC source and capture instrument instance. An instrument instance, whether source or capture, has four ports, the first of which is the compatibility port. The compatibility port can be programmed using BBAC syntax or TurboAC syntax. The other three ports of the instrument instance can only be programmed using TurboAC syntax.
CW	Continuous Waveform Source. CW is a capability that increases the TurboACSource bandwidth by allowing a synthesizer to be routed through the instrument delivery path.
DGS	Device ground sense
DIB LO	A user supplied signal that can be used as an alternative to the Standard or the Optional CW as the capture instrument's local oscillator on the TurboAC board.
DIB Source	A user supplied source signal that can output to the TurboACSource ports.
direct capture mode	Allows capture and process of signals in a bandwidth from DC to 15MHz.

down conversion	A mixing technique used in a capture instrument to shift the tones being captured to another frequency. The basic capture instrument has a bandwidth of 15MHz. Used normally, a capture would contain tones between DC and 15MHz. Using the down conversion circuitry, a capture can be made of 10MHz sections of bandwidth of signals at higher frequencies (for example, the 10MHz band between 65 - 75MHz).
Gsps	Gigasamples per second
HSD	High Speed Digital
ISL	Instrument Sync-Link
ksps	Kilosamples per second
Msps	Megasamples per second
Optional CW	An optional high frequency continuous waveform source that can be used with the TurboAC instrument. This frequency synthesizer can be of any brand. When connected to the TurboAC instrument, this frequency synthesizer would provide the optional CW signal for the source channel, and the optional LO signal for the capture channel. IG-XL provides programming and calibration support for the optional CW generator.
Optional LO	Alternate term for the Optional Frequency Synthesizer when used as a LO.
port	The set of pins that carries a signal to or from a location on the DIB. For the TurboACSource a port is composed of three pins: Pos, Neg and Ref. Each instrument instance has four capture ports labeled A, B, C, and D.
PPMU	Pin Parametric Measurement Unit
PSet	Parameter set
signal cancellation	A technique to suppress a tone in a captured signal. The signal cancellation technique requires a signal from the source instrument instance that is on the same board as the capture instrument instance. A signal cancellation application typically requires a pattern to control the relative phase between the canceling signal and the captured signal.
SMEM	Source Memory

Standard CW	The default frequency synthesizer that is part of a TurboAC instrument. The PTS provides the continuous waveform signal for the source channel's CW input.
Turbo-Base channel card	The TurboACSource back end circuitry, including the AWG and capture hardware.
Turbo-Frame channel card	The TurboACSource analog front end circuitry, including: - Filters - Down convertor - Signal delivery - Signal Combination - CW DC offset

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About the TurboACSource : TurboAC Source Software Licensing Requirements

TurboAC Source Software Licensing Requirements

TurboAC instrument hardware can support multiple frequencies across source and capture instruments. However, IG-XL enforces a software licensing scheme that constrains instrument functionality to a specified frequency limit across source and capture instruments. The TurboAC instrument can be configured with one of the following licenses:

Full License	A full license enables: • Full specification performance and functionality for two source and two capture instruments • AWG mode at 15MHz Optional CW is enabled by a separate license.
Single License	A single license enables: • TurboAC Frame card including 1 MX321 Turbo Channel card with Riders • 1 PTS310 synthesizer in the Support or Expansion cabinet • A Synthesizer Interface Module within the same u-space as the PTS

Audio License

- AWG mode at 15MHz

Optional CW source is enabled by a separate license.

An audio license enables:

- One capture and one source instrument
- AWG mode at 3MHz

CW source and the down conversion features are not available with this license.

License Error Handling

IG-XL checks that the correct license and configuration are on the test system. If the license and configuration do not match, a run-time error is returned at validation time. The following table lists how IG-XL handles license and configuration mismatches:

License and Configuration Mismatch Handling			
Configuration	Full License	Single License	Audio License
Full Configuration	No Error.	Error: Turbo-CC-2 or PTS-2 is not licensed.	Error: PTS-1 or 2 is not licensed.
Single Configuration	Error: Trying to use Non-existing Turbo-CC-2 or PTS-2.	No Error.	Error: PTS-1 is not licensed. Error: Trying to use Non-existing Turbo-CC-2 or PTS-2.
Audio Configuration	Error: Trying to use Non-existing PTS-1 or 2.	Error: Turbo-CC-2 is not licensed. Error: Trying to use Non-existing PTS-1 (or 2).	No Error.

Related Topics

For more information, see IG-XL Licensing.

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TurboACSource Debug Display

TurboACSource Debug Display

The TurboACSource debug display is an interactive diagram of the TurboACSource hardware and signal path, which shows the instrument’s settings and connections in the context of a site. You can change the values, options, and settings in the display and run your test program to see the effects of the changes.

The following topics are available:

- Starting the TurboACSource Debug Display
- TurboACSource Debug Display Interface
- Using the TurboACSource Debug Display

This section describes features specific to the TurboACSource debug display. For more information on using debug displays, see the following:

- Using Debug Displays in TDE
- Producing Code from Debug Displays

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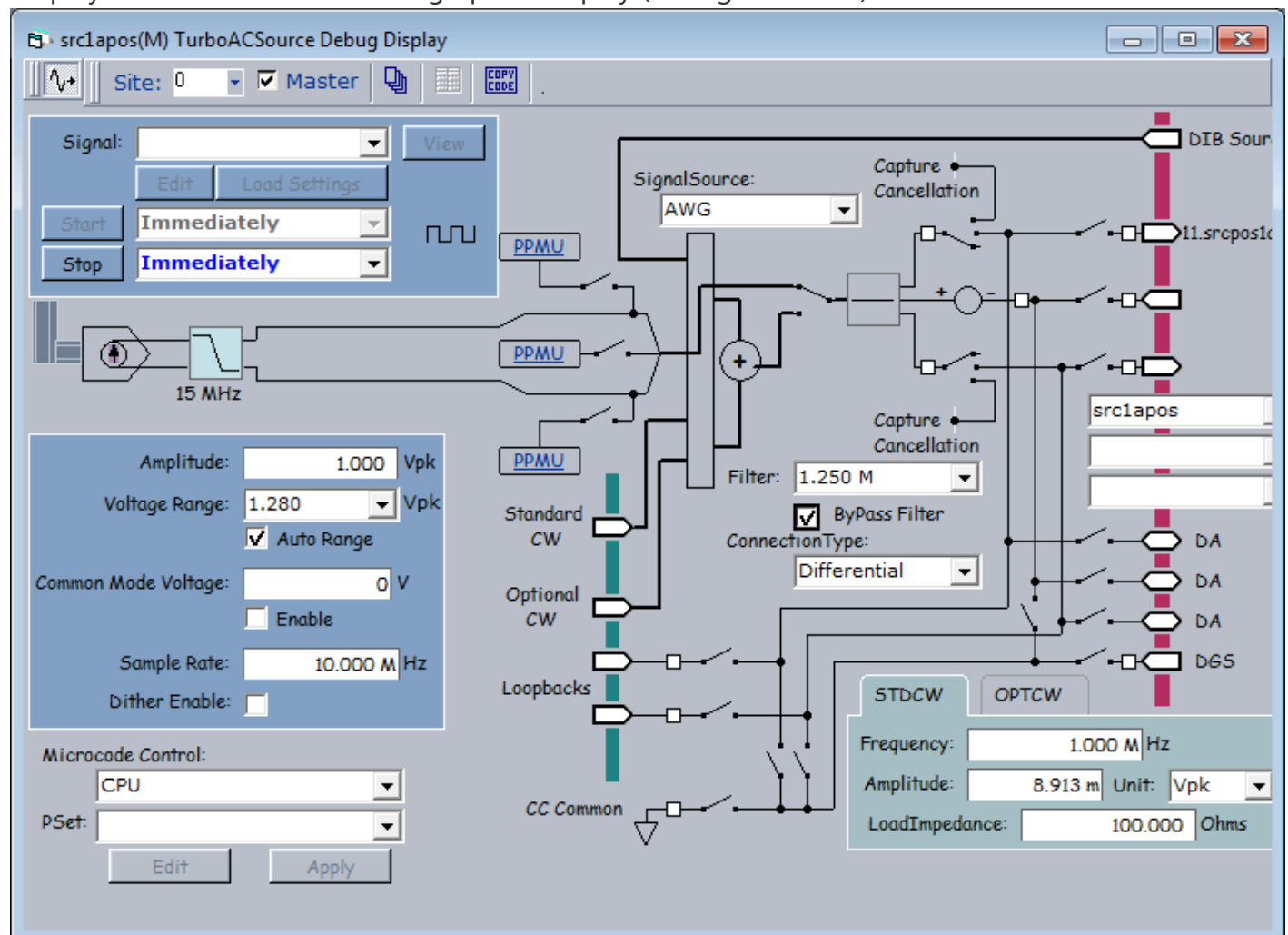
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TurboACSource Debug Display : Starting the TurboACSource Debug Display

Starting the TurboACSource Debug Display

To launch the TurboACSource Debug Display, first validate a test program that includes TurboACSource pins. Then, on the IG-XL Tools tab in the Launch group, select Debug Displays>TurboACSource to bring up the display (see figure below).



TurboACSource Debug Display

If you start a debug display before the job has been validated, an error appears in the display saying that the program has not been validated. If you open a display for an instrument that is in the configuration file but not in the channel map, the display shows a message that no instruments have been found.

Once the job is validated, the display shows the current hardware settings. When you run to a breakpoint in the flow or in the VBT code, or as you step through the flow or code, IG-XL updates the display to show any changes to the settings.

For information on other methods for starting debug displays, see [Starting Debug Displays](#) in the Test Debug Environment (TDE) section.

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TurboACSource Debug Display : TurboACSource Debug Display Interface

TurboACSource Debug Display Interface

The following sections describe the interface for the TurboACSource debug display:

- [Source Signal](#)
- [TurboACSource Parameters](#)
- [Microcode Control](#)
- [Continuous Wave Source](#)
- [Signal Type and Filter Settings](#)

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TurboACSource Debug Display : TurboACSource Debug Display Interface : Source Signal

Source Signal

This section of the display includes input boxes and buttons that select which signal to source and where the source instrument receives the start and stop commands from.

You can use the View button to preview the signal to be sourced in a scope display. Note that most of the fields and buttons only become active once a signal is specified in the Signal parameter.

Signal:

View

Edit

Load Settings

Start

Immediately

Stop

Immediately



Source Signal Block

The parameters and buttons that control the sourcing of signals are:

Signal	The names of the defined signals. If a default signal has been defined, the default signal’s name appears automatically in this field. If the menu contains no signal names, no signals were previously defined in VBT.
View	Brings up the WaveScope tool and plots the sourced signal segment on the WaveScope display. This allows you to visually preview and analyze the signal that is to be sourced. When you are running the program on a test system, WaveScope displays the samples and scales them by the amplitude. When working offline, the WaveScope displays the ideal waveform.

Edit	Brings up the PSet and Signals Editor (PSSE) with the selected signal loaded. This control is grayed out if no signals are available. See PSet and Signals Editor in the TDE section.
Load Settings	Loads the signal's timing settings as specified on the Mixed Signal Timing sheet, as well as all of the instrument's settings (for example, voltage range, amplitude, and offset).
Start	Starts sourcing the signal for the specified signal according to the action command selected in the menu next to the Start button. If the menu displays At End the instrument sources the named signal in SMEM and repeats it continuously until one of the following happens: • A STOP microcode is executed • The instrument is reset • Another signal is started If the menu displays Immediately, the instrument starts sourcing (on the next available logical analog clock) the named signal in SMEM and repeats it continuously until one of the following happens: • A STOP microcode is executed • The instrument is reset • Another signal is started If the menu displays Once, the instrument starts sourcing (on the next available logical analog clock) the named signal in SMEM and sources it exactly once. If the menu displays At End, the instrument sources the named signal in SMEM and sources it exactly once.
Stop	Stops sourcing according to the action command selected in the menu next to the Stop button. If the menu displays At End, the instrument stops sourcing the currently sourced signal at the end of the signal.



If the menu displays Immediately, the instrument stops sourcing the currently sourced signal immediately. Immediately does not impact a signal that is running once.

This a sourcing indicator that moves when the source is sourcing on the tester. This feature only works online.

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TurboACSource Debug Display : TurboACSource Debug Display Interface : TurboACSource Parameters

TurboACSource Parameters

This block includes input boxes that allow you to select the voltage and amplitude settings of the output signal. You can also set the common mode voltage to center the AC signal around a common mode voltage, or you can set the Auto Range feature to automatically adjust the voltage range to source the signal within the largest dynamic range. At the bottom of the block is a check box to enable the addition of dither to a signal.

Amplitude:1.000Vpk

Voltage Range:1.280Vpk

☒ Auto Range

Common Mode Voltage:0V

☐ Enable

Sample Rate:10.000MHz

Dither Enable:☐

Voltage Range and Amplitude Block

The parameters and buttons that control the voltage and amplitude of signals are:

Amplitude	Sets the amplitude of the signal as it will be seen by the DUT. AWG mode.
Voltage Range	Sets the voltage range of the signal. The menu provides the available voltage ranges that can be programmed. AWG mode.
Auto Range	When selected, auto range mode is active. As the amplitude changes, the instrument automatically changes the voltage range so that it has the highest signal fidelity by using the entire dynamic range. Using Auto

	Range results in applying the smallest voltage range that can support the programmed amplitude. When auto range is deactivated, the voltage range is left at the last voltage range specified. AWG mode.
Common Mode Voltage	Sets the common mode voltage around which the AC signal is centered.
Enable	Enables Common Mode Voltage.
Sample Rate	Represents the number of samples to source per second. AWG mode.
Dither Enable	When checked, enables the source to add dither to the signal. AWG mode.

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TurboACSource Debug Display : TurboACSource Debug Display Interface : Microcode Control

Microcode Control

This section of the display includes controls to select the origin of the pattern microcode start command and a PSet to be applied later.

Microcode Control:

CPU

PSet:

Edit

Apply

Microcode and PSet Block

The parameters and buttons that control microcode settings are:

Microcode Control	Specifies where the pattern microcode start command, which starts the sourcing of the signal, originates. The menu provides the following options: - CPU: From the host computer - External Trigger: From the DIB - Slaved External Trigger: From another instrument
PSet	Selects a parameter set (PSet). However, it does not apply the setting specified in the PSet to hardware. The PSet is only applied to hardware when you click the Apply button.

Edit	Brings up the PSet and Signals Editor (PSSE) with the selected PSet loaded. This control is grayed out if no PSets are available. See PSet and Signals Editor in the TDE section.
Apply	Applies the PSet specified in the PSet field to the sourced signal.

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TurboACSource Debug Display : TurboACSource Debug Display Interface : Continuous Wave Source

Continuous Wave Source

This section of the display includes controls to set parameters for the standard or optional CW source. Click a tab to bring up the parameters for the CW source you are interested in.

STDCW

OPTCW

Frequency:

1.000 M

Hz

Amplitude:

8.913 m

Unit:

Vpk

LoadImpedance:

100.000

Ohms

Continuous Wave Source Parameters

The parameters that control the continuous wave source are:

Frequency	Sets the frequency for the continuous wave source.
Amplitude	Sets the amplitude of the signal as it will be seen by the DUT in vpk or vpp.
LoadImpedance	Sets the DUT load for the continuous wave source.

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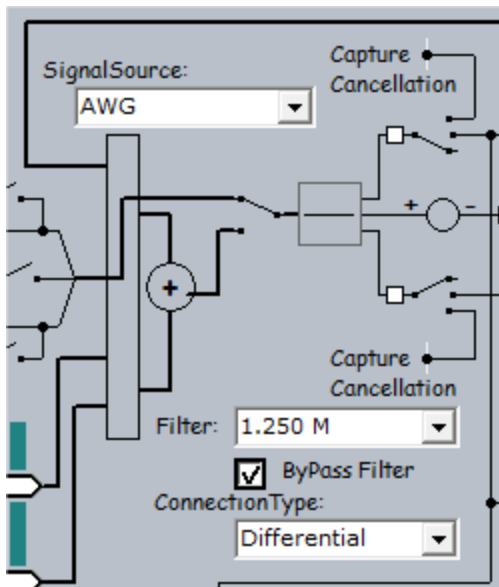
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TurboACSource Debug Display : TurboACSource Debug Display Interface : Signal Type and Filter Settings

Signal Type and Filter Settings

Several controls in the display allow you to set the source type and output type and the filter.



Signal Type and Filter Settings

The parameters that control signal type, output, and filter are:

SignalSource

Sets the source to be used for the TurboACSource. When you select from the menu, the debug display functional diagram updates to reflect the connection. Choose from the following:

- **AWG**: AWG signal source only.
- **StdCW**: Standard CW source only.

- [OptCW](#): Optional connected CW source only.
- [DIBSrc](#): DIB source only.
- [AWG + StdCW](#): AWG combined with standard CW source.
- [AWG + DIBSrc](#): AWG combined with DIB source.
- [AWG + OptCW](#): AWG combined with optional connected CW source.
- [StdCW + DIBSrc](#): Standard CW source combined with DIB source.
- [StdCW + OptCW](#): Standard CW source combined with optional connected CW source.
- [OptCW + DIBSrc](#): Optional connected CW source combined with DIB source.

Capture Cancellation Capture cancellation is set up using the TurboACCapture.

ConnectionType Sets if the Source output is single-ended or differential for the TurboACSource. Select from the menu:

- [Differential](#): Sets the output to differential mode.
- [Single-Ended](#): Sets the output to single-ended mode.

Filter Sets a filter size to be used if the filter is being used.

ByPass Filter Sets the filter circuitry to be bypassed. Deselecting the check box uses the filter setting specified in the [Filter](#) menu.

PPMU Click this button to open the PPMU debug display for the selected pin. See [PPMU Debug Display](#).

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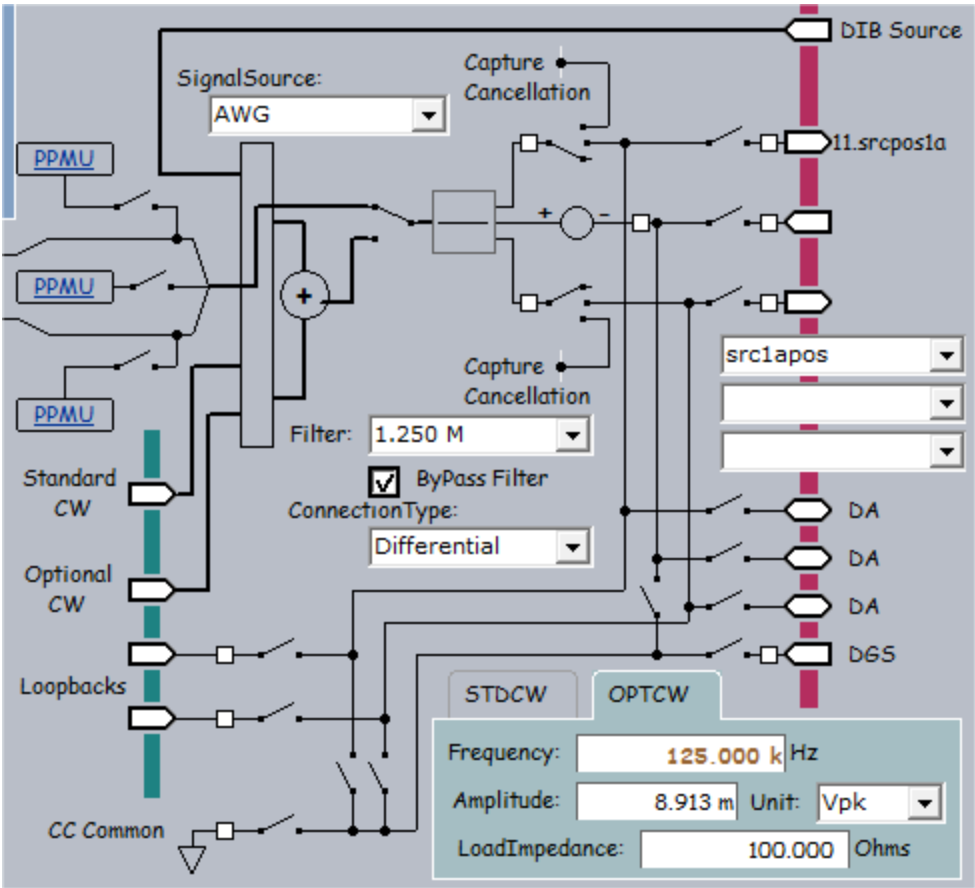
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TurboACSource Debug Display : Using the TurboACSource Debug Display

Using the TurboACSource Debug Display

The figure shows the connections matrix on the TurboACSource debug display. This matrix shows all possible TurboACSource channel connections to available resources.



TurboACSource Debug Display Connections Matrix

The following topics describe the interface behavior and the actions that you perform to make channel and resource connections in the TurboAC debug display.

- Opening and Closing a Relay Path

- [Highlighting a Closed Relay Path](#)

- [Determining a Local or Indirect Channel Connection](#)

The following topics describe the resources that the TurboACSource channels can be connected to:

- [Connecting DUT Pins](#)

- [Connecting DIB Access Channels](#)

- [Connecting Device Ground Sense](#)

- [Connecting the Auxiliary Pin](#)

- [Connecting the Internal Analog Loopback Path](#)

- [Connecting a PPMU](#)

- [Connecting Channel Card Common](#)



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TurboACSource Debug Display : Using the TurboACSource Debug Display : Opening and Closing a Relay Path

Opening and Closing a Relay Path

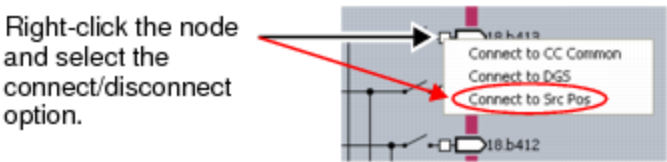
The TurboACSource debug display simulates channel connections to the DUT and other resources. Graphically, these connections are represented by open and closed relays along the connection path. Clicking a relay does not immediately open or close that relay. It does, however, open a menu that lists the connection options for that relay. You can then select the desired connection action and destination. Note that the relays associated to the Auxiliary and PPMU connections cannot be connected from the TurboACSource debug display (see [Connecting the Auxiliary Pin](#) and [Connecting a PPMU](#)).

To open or close a relay path:

1.

Right-click the node along the signal path. A menu appears and lists the connect or disconnect options for that path.
2.

Select the connection destination from the menu. The relay opens or closes to make or break the connection.



Connect/Disconnect Channels and Resources Menu

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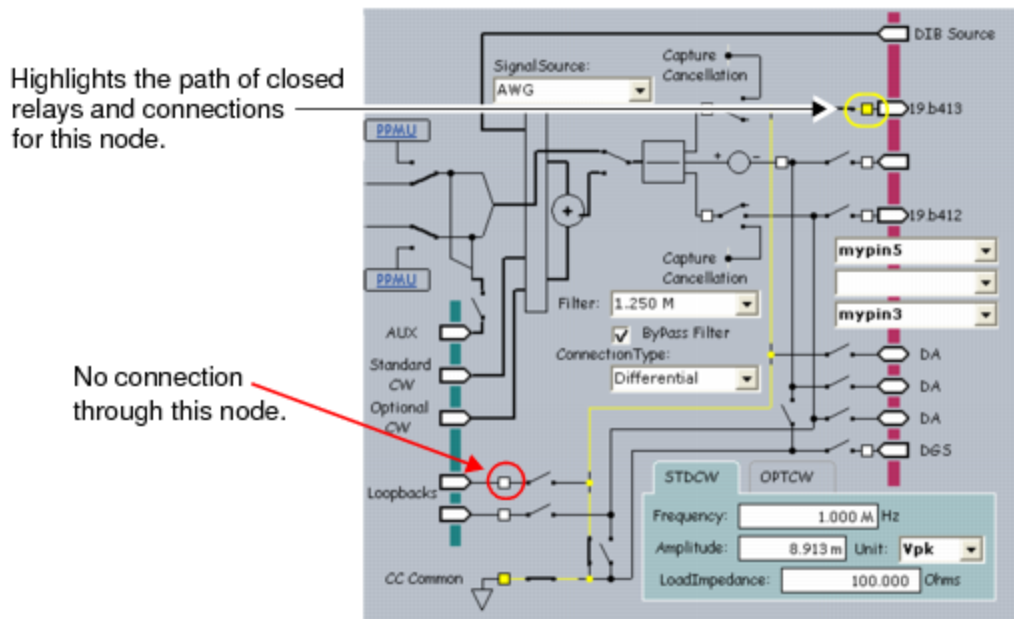


TurboACSource Debug Display : Using the TurboACSource Debug Display : Highlighting a Closed Relay Path

Highlighting a Closed Relay Path

To determine which resources are connected to a specific TurboACSource channel, you can highlight a connection's relay path.

The figure shows that you can highlight a closed relay path by left-clicking and holding the node along the closed relay path. The path of closed relays and connections that run through that node are then highlighted in yellow. If no connection goes through that node, the node turns red.



Tracing a Closed Relay Path



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TurboACSource Debug Display : Using the TurboACSource Debug Display : Determining a Local or Indirect Channel Connection

Determining a Local or Indirect Channel Connection

The debug display's connection matrix shows the connection paths related to TurboACSource channels. These connections can result from connections made on this display or those made on another instrument's debug display. For example, if a TurboACSource pin is connected to a PPMU or the auxiliary connection, the connection is made in the PPMU or primary instrument's display, and the resulting connection appears in both the TurboACSource and the PPMU/primary instrument's debug display.

You can determine which TurboACSource channel connections were made locally and which were made by another instrument display. When you click the DUT port of a channel, a menu appears and lists all pins behind a particular instrument channel. A check mark next to the pin name indicates that the connection is closed from the TurboACSource instrument to the DUT. An asterisk indicates that the connection was made as a result of another connection.

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TurboACSource Debug Display : Using the TurboACSource Debug Display : Connecting DUT Pins

Connecting DUT Pins

The TurboACSource debug display allows you to connect an instrument channel and other resources to the DUT pins specified in the test program’s channel map.

Starting the debug display automatically reads in the pins and lists them in the DUT port menus.

The DIB channel string is also read into the debug display and is shown beside the instrument channel.

To connect an instrument channel or resource to a DUT pin:

- | | |
|----|---|
| 1. | Right-click the DUT node. |
| 2. | Select the destination resource or channel from the menu. |

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TurboACSource Debug Display : Using the TurboACSource Debug Display : Connecting DIB Access Channels

Connecting DIB Access Channels

DIB access is an instrument feature that permits a DUT pin to be connected to one instrument board through another instrument board. It allows you to connect multiple resources on different boards to a single pin.

Since DIB access involves connecting instruments, it also means coordinating connections between two instrument debug displays. To distinguish one instrument from another, one is called the primary instrument and the other is the secondary. The primary instrument is the one that uses another instrument’s DIB access path. The secondary instrument is the one whose path is used by the primary instrument.

In making a DIB access connection, you can only close that connection from the primary instrument’s debug display.

Debug Display DIB Access Connection Example

The following instructions show how to make a DIB access connection between two instrument debug displays. In this case, a TurboACSource is using a TurboACCapture’s DIB access path. That means that the TurboACSource is the primary instrument and the TurboACCapture is the secondary instrument.

To make a debug display DIB access connection for this example:

- | | |
|----|--|
| 1. | Start the TurboAC Source debug display. |
| 2. | Select the pin in the DUT Pos menu specified as a DIB access pin. In this case, it is cap1pos. |
| 3. | Right-click the SrcPos node and select Connect to Src Pos from the menu. |

4. [Start the TurboACCapture debug display.](#)
5. [Select the pin \(cap1pos\) from the DUT Pos menu.](#)

The DIB Access path relays automatically close and the connection is complete.

Since the TurboACCapture is the secondary instrument in this example, you cannot disconnect the DIB access path from within its environment. You must return to the primary instrument, the TurboACSource, to open the DIB access connections.

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TurboACSource Debug Display : Using the TurboACSource Debug Display : Connecting Device Ground Sense

Connecting Device Ground Sense

Device ground sense (DGS) is a signal that can be connected to the ground of the DUT. DGS is used as a lowside reference for the PPMUs. Normally, you do not connect source signals to the DGS lines because that will improperly cause current to flow in the sense line.

To connect a DUT pin or instrument channel to the DGS:

1.

Right-click the DGS node.
2.

Select the destination channel from the menu.

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TurboACSource Debug Display : Using the TurboACSource Debug Display : Connecting the Auxiliary Pin

Connecting the Auxiliary Pin

The auxiliary connection to the TurboACSource allows another instrument to connect to the TurboAC.

When the auxiliary connection is closed, a TurboACSource is considered a secondary instrument, and the primary instrument is using the TurboAC instrument. The auxiliary connections on the debug display are read-only and show whether the auxiliary connections are open or closed. To open or close the auxiliary connection, make the connection from the primary instrument. Once the auxiliary connection is closed, no other connections on the TurboACSource can be made. When the auxiliary connection is closed, the Src Pos pin is connected to AUX Pos, Src Ref is connected to CCC, and Src Neg pin is connected to AUX Neg.

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TurboACSource Debug Display : Using the TurboACSource Debug Display : Connecting the Internal Analog Loopback Path

Connecting the Internal Analog Loopback Path

The loopback connection is provided for debugging. It allows you to connect a TurboACCapture directly to a TurboACSource, allowing it to make a measurement of the sourced waveform. This connection allows TurboACSource1 to loop back to TurboACCapture1 or TurboACSource2 to loop back to TurboACCapture2. You can independently control the Pos and Neg channels. (For example, you can choose to loop back Positive, Negative, or Both.) For details on see Internal Loopback Circuitry.

To connect the TurboACCapture to the TurboACSource for loopback purposes:

1. Right-click the Loopback Positive node and select Connect to Src Pos from the menu.
2. Right-click the Loopback Negative node and select Connect to Src Neg from the menu.
3. Right-click the Source Reference node and select CCC from the menu.
4. Open the TurboACCapture debug display.
5. Right-click the Loopback Positive node and select Connect to Cap Pos from the menu.
6. Right-click the Loopback Negative node and select Connect to Cap Neg from the menu.

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TurboACSource Debug Display : Using the TurboACSource Debug Display : Connecting the Internal Cross Channel Analog Loopback Path

Connecting the Internal Cross Channel Analog Loopback Path

The cross channel loopback connection is provided for debugging. It allows you to connect a TurboACCapture directly to a TurboACSource, allowing it to make a measurement of the sourced waveform.

This connection allows TurboACSource1 to loop back to TurboACCapture2 or TurboACSource1 to loop back to TurboACCapture2. You can independently control the Pos and Neg channels. (For example, you can choose to loop back Positive, Negative, or Both.) For details on see Internal Loopback Circuitry.

To connect the TurboACCapture to the TurboACSource for loopback purposes:

1.

Right-click the Loopback Positive node and select Connect to Src Pos from the menu.
2.

Right-click the Loopback Negative node and select Connect to Src Neg from the menu.
3.

Right-click the Source Reference node and select CCC from the menu.
4.

Open the TurboACCapture debug display.
5.

Right-click the Loopback Positive node and select Connect to Cap Pos (Cross Channel) from the menu.
6.

Right-click the Loopback Negative node and select Connect to Cap Neg (Cross Channel) from the menu.

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TurboACSource Debug Display : Using the TurboACSource Debug Display : Connecting a PPMU

Connecting a PPMU

You can connect each TurboACSource channel to a Pin Parametric Measurement Unit (PPMU) circuit for performing continuity tests.

Although the TurboACSource display shows the channel connections to the PPMU circuits, PPMU connections must be made in the PPMU display, not the TurboACSource display.

To connect a TurboACSource DUT pin (Pos, Neg, or Ref) to a PPMU circuit:

1.



Click the PPMU link on the TurboACSource display based on the TurboACSource channel to be connected. This action starts the PPMU debug display.
2.

Select the pin to be connected to the PPMU from the Pin menu.
3.

Click the relay to close the connection.

For more information, see PPMU Debug Display.

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TurboACSource Debug Display : Using the TurboACSource Debug Display : Connecting Channel Card Common

Connecting Channel Card Common

You can connect a DUT pin to the channel card common (CCC) ground through the positive, reference, or negative channels.

To connect a DUT pin or instrument channel to the CCC ground:

- | | |
|----|---|
| 1. | Right-click the CC Common node. |
| 2. | Select the destination channel from the menu. |

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TurboACSource Programming

TurboACSource Programming

This section describes programming of a TurboACSource to source a waveform to a device.
The following topics are available:

- [Simulated Configuration Files](#)
- [TurboACSource DataTool Programming](#)
- [TurboACSource Connections](#)
- [Visual Basic for Test](#)
- [TurboACSource Reset Values](#)
- [TurboACSource Pattern Control](#)
- [Using TurboAC in Concurrent Testing](#)

For more information, see:

- [TurboACSource](#) (VBT Syntax Reference)
- [TurboACSource Examples](#)

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TurboACSource Programming : Simulated Configuration Files

Simulated Configuration Files

You can use TurboAC Source offline by forcing IG-XL to map the TurboAC instrument. Do this by creating a SimulatedConfig.txt file and placing the file in any of the following locations:

- Folder containing the IG-XL workbook
- \tester folder: C:\Program Files (x86)Teradyne\IG-XL\<version>\tester
- \bin folder: C:\Program Files (x86)\Teradyne\IG-XL\<version>\bin

The following SimulatedConfig.txt file maps the TurboAC instrument in slots 11 and 23:
UltraFLEX IG-XL hardware configuration
This file contains an example hardware configuration, and
is used as the default offline.

```
<Version>
1
</Version>

<TEST_HEAD TH 1>
#slot[subslot] Type      idprom (type serial rev company)

-1.0      dib      239-001-00 0000000 0000-A 0000

1.0      HSD-800    979-219-00 0000000 0000-A 0000
```

	HSD-800	956-361-00 0000000 0000-A 0000
6.0	DC-30	805-002-00 0000000 0000-A 0000
	DC-30	949-901-00 0000000 0000-A 0000
3.0	GigaDigHD	979-999-00 0000000 0000-A 0000
	GigaDigHD	956-436-00 0000000 0000-A 0000
11.0	Turbo-Frame	974-213-01 03017D1A 0549-B 5445
	aka: 974-213-00	
	Turbo-Frame	939-367-01 03014072 0541-B 5445
	Turbo-CC-1	939-321-00 030172BE 0544-C 5445
	PowerBoard	939-364-00 03019B50 0528-B 5445
	Turbo-CC-2	939-321-00 030172C8 0544-C 5445
11.1	Turbo-SIM	974-228-01 03014213 0546-B 5445
	aka: 974-228-00	
	Turbo-SIM	939-366-01 0301570F 0542-B 5445
	aka: 939-366-00	
12.0	BBAC-15	979-214-00 0000000 0000-A 0000
	BBAC-15	956-375-00 0000000 0000-A 0000
23.0	Turbo-Base	974-215-00 03013B33 0549-D 5445
	Turbo-Base	956-375-14 03013A23 0549-D 5445
	aka: 956-078-00	
	Turbo-BaseRider	949-987-02 0301668A 0451-D 5445
	PowerBoard	949-984-04 03010C72 0502-D 5445
	aka: 949-984-00	
24.0	SupportBoard	974-220-12 0000000 0000-A 0000
	SupportBoard	956-354-00 0000000 0000-A 0000

</TEST_HEAD TH 1> t

To use the TurboAC instrument offline, the SimulatedConfig.txt file must include every board needed in the IG-XL workbook. The SimulatedConfig.txt file represents the state of the offline tester. To use the TurboAC instrument online, you need not edit any files. IG-XL automatically maps all boards in the tester.

For more information, see [Tester Configuration](#).

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TurboACSource Programming : TurboACSource DataTool Programming

TurboACSource DataTool Programming

This section describes how to program several DataTool worksheets to handle basic TurboACSource parameters. It includes the following topics:

- | | |
|---|--|
| • | Programming the Pin Map |
| • | Programming the Channel Map |
| • | Creating a Wave Definition and Mixed Signal Timing Set in AWG Mode |

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[TurboACSource Programming](#) : [TurboACSource DataTool Programming](#) : Programming the Pin Map

Programming the Pin Map

A pin map describes the device pins by assigning a pin type to each device pin name. The recommended Type selection for the TurboACSource instrument is:

Analog	This type is used for the TurboACSource instrument.
--------	---

For more information, see [Pin Map Sheet](#) in the DataTool section.

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TurboACSource Programming : TurboACSource DataTool Programming : Programming the Channel Map

Programming the Channel Map

A Channel Map sheet assigns individual device pin names to test system channels and power supplies. For the TurboACSource instrument, you must select a channel type from the Type column menu for each pin or pin group. Valid TurboACSource channel types are:

BBACSrcNeg	TurboACSource Negative Pin when in BBAC compatible mode.
BBACSrcPos	TurboACSource Positive Pin when in BBAC compatible mode.
BBACSrcRef	TurboACSource Reference Pin when in BBAC compatible mode.
TurboACSrcNeg	TurboACSource Negative Pin.
TurboACSrcPos	TurboACSource Positive Pin.
TurboACSrcRef	TurboACSource Reference Pin.

See [Specifying Tester Channels](#) in the DataTool section for information on determining the tester channel information value.The Signal naming conventions are:

- <slot #>.srcpos<channel 1 a - d> (positive pin, for example: 11.srcpos1a)
- <slot #>.srcpos<channel 2 a - d> (positive pin, for example 11.srcpos2c)
- <slot #>.srcneg<channel 1 a - d> (negative pin, for example 11.srcneg1b)
- <slot #>.srcneg<channel 2 a - d> (negative pin, for example 11.srcneg2d)
- <slot #>.srcref<channel 1 a - d> (reference pin, for example 11.srcref1b)
- <slot #>.srcref<channel 2 a - d> (reference pin, for example 11.srcref2d)

For more information, see [Channel Map Sheet](#) in the DataTool section.

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TurboACSource Programming : TurboACSource DataTool Programming : Creating a Wave Definition and Mixed Signal Timing Set in AWG Mode

Creating a Wave Definition and Mixed Signal Timing Set in AWG Mode

A TurboACSource instrument generates and delivers analog signals of a specific shape, frequency, and amplitude to the DUT. These signals start as wave definitions, which define the basic shape of the signal. Then a timing solution is added to the waveform definition to create a timing set for the test. The TurboACSource instrument uses the timing set to create a reusable signal to which various amplitudes can be applied.

The following topics provide information on creating a wave definition and a timing set:

- | | |
|---|---|
| • | Setting Up the Wave Definitions Sheet |
| • | Creating a Timing Set Using the Mixed Signal Workshop |
| • | Calculating Source Timing |
| • | Source Timing Example: 100Hz Sine Wave |
| • | Source Timing Example: 1.5MHz in High Frequency Mode |
| • | Source Timing Example: 13MHz in High Frequency Mode |

Setting Up the Wave Definitions Sheet

A Wave Definitions sheet contains the wave shape information for waveform definitions that were created using the WaveDesigner. A Wave Definitions sheet stores the wave definition information,

and the test program accesses the information when the test is run.

For more information, see [About WaveDesigner](#) in the WaveDesigner section.

Creating a Timing Set Using the Mixed Signal Workshop

The Mixed Signal Workshop (MSW) tool helps you to build reusable mixed-signal timing solutions that meet coherence requirements.

The WaveDesigner tool creates wave definitions, which are independent of timing and instrument specific information. The MSW uses these wave definitions and then applies the instrument specific timing information necessary to generate samples.

The output of the MSW is a mixed-signal timing set which contains the signal's structural characteristics and the details of the clocking solution that you created. Timing sets are stored on a Mixed Signal Timing sheet in the IG-XL workbook and become a reusable timing solution for future tests.

For more information, see [Mixed Signal Workshop](#).

Calculating Source Timing

Source timing solutions must adhere to the TurboACSource timing rules.

The minimum F_s allowed is:

$$(1) \quad F_s \geq \text{Maximum [5 kHz or } 20 \times F_i]$$

Where F_i is the highest frequency of interest.

The minimum N allowed is:

$$(2) \quad N \geq \text{Maximum} \left[20 \text{ or } \left(\text{ceiling} \left(F \frac{s}{100 \text{ Msps}} \right) \times 8 \right) \right]$$

Source Timing Example: 100Hz Sine Wave

The following example shows the source timing solution calculation for a 100Hz sine wave.

$$F_s \geq \max [5\text{kHz} \parallel 20 \times F_i]$$

$$F_s \geq \max [5\text{kHz} \parallel 20 \times 100\text{Hz}]$$

$$F_s \geq \max [5\text{kHz} \parallel 2\text{kHz}]$$

$$F_s \geq 5\text{kHz}$$

$$N \geq \max [20 \parallel \text{ceiling}(F_s/100\text{Msps}) \times 8]$$

$$N \geq \max [20 \parallel \text{ceiling}(5\text{ksps}/100\text{Msps}) \times 8]$$

$$N \geq \max [20 \parallel 8]$$

$$N \geq 20$$

With $F_s = 5\text{kHz}$ and $M = 1$:

$$N = M \times F_s/F_i$$

$$N = 1 \times 5\text{kHz}/100\text{Hz}$$

$$N = 50 \text{ meets minimum requirement}$$

Therefore, a valid timing solution for a 100Hz sine wave is:

$$F_s = 5\text{kHz}$$

$$N = 50$$

$$M = 1$$

Source Timing Example: 1.5MHz in High Frequency Mode

The following example shows the source timing solution calculation for a 1.5MHz sine wave.

$$F_s \geq \max [5\text{kHz} \parallel 20 \times F_i]$$

$$F_s \geq \max [5\text{kHz} \parallel 20 \times 1.5\text{MHz}]$$

$$F_s \geq \max [5\text{kHz} \parallel 30\text{MHz}]$$

$$F_s \geq 30\text{MHz}$$

$$N \geq \max [20 \parallel \text{ceiling}(F_s/100\text{Msps}) \times 8]$$

$$N \geq \max [20 \parallel \text{ceiling}(30\text{Msps}/100\text{Msps}) \times 8]$$

$$N \geq \max [20 \parallel 8]$$

$$N \geq 20$$

With $F_s = 30\text{MHz}$ and $M = 1$:

$$N = M \times F_s/F_i$$

$$N = 1 \times 30\text{MHz}/1.5\text{MHz}$$

$N = 20$ meets the minimum requirement

Therefore, a valid timing solution for a 1.5MHz sine wave is:

$$F_s = 30\text{MHz}$$

$$N = 20$$

$$M = 1$$

Source Timing Example: 13MHz in High Frequency Mode

The following example shows the source timing solution calculation for a 13MHz sine wave.

$$F_s \geq \max [5\text{kHz} \parallel 20 \times F_i]$$

$$F_s \geq \max [5\text{kHz} \parallel 20 \times 13\text{MHz}]$$

$$F_s \geq \max [5\text{kHz} \parallel 260\text{MHz}]$$

$$F_s \geq 260\text{MHz}$$

$$N \geq \max [20 \parallel \text{ceiling}(F_s/100\text{Msps}) \times 8]$$

$$N \geq \max [20 \parallel \text{ceiling}(260\text{Msps}/100\text{Msps}) \times 8]$$

$$N \geq \max [20 \parallel 24]$$

$$N \geq 24$$

With $F_s = 260\text{MHz}$ and $M = 1$:

$$N = M \times F_s/F_i$$

$$N = 1 \times 260\text{MHz}/13\text{MHz}$$

$N = 20$ does not meet the minimum requirement

So, change M to 2. With $F_s = 260\text{MHz}$ and $M = 2$:

$$N = M \times F_s/F_i$$

$$N = 2 \times 260\text{MHz}/13\text{MHz}$$

$N = 40$ meets the minimum requirement

Therefore, a valid timing solution for a 13MHz sine wave is:

Fs = 260MHz
N = 40
M = 2

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TurboACSource Programming : TurboACSource Connections

TurboACSource Connections

The TurboACSource can connect to the DUT, CC Common, DGS, and Capture Cancellation and to a TurboACCapture in a Loopback. You can also determine to connect the positive pin, negative pin, reference pin, or all three pins.

You make connections using VBT syntax or the TurboACSource debug display. The following topics on making connections are available:

- Connecting to the DUT Through TurboACSource Port
- Connecting to the Channel Map Specified DIB Site
- Connecting to a Selected Port
- Connecting Individual Pins to a Port
- BBAC Compatible Connections
- Making a Differential Connection to Capture Cancellation Circuitry
- Making a Single-Ended Connection to Capture Cancellation Circuitry
- Connecting a DUT Pin to CC Common

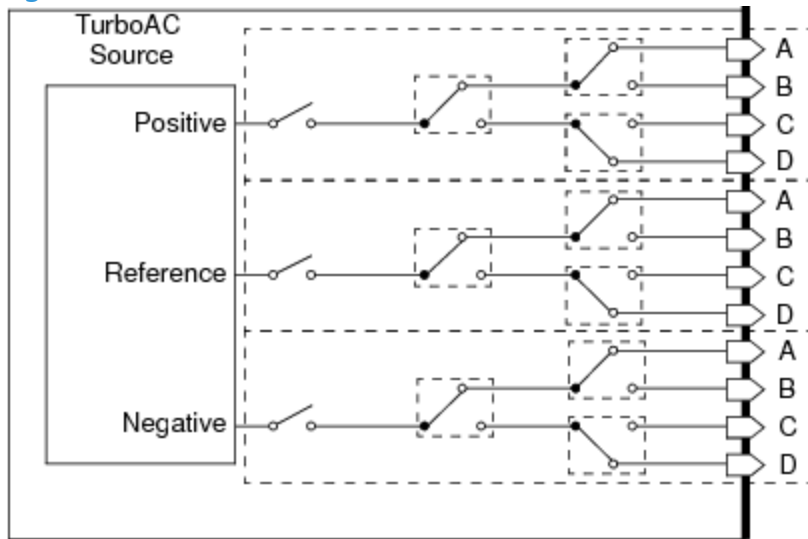
- [Connecting to DGS](#)
- [Connecting the Loopback Circuitry](#)
- [Connecting the Cross Channel Loopback Circuitry](#)
- [Connecting to a PPMU](#)
- [DIB Access Connections](#)

turboacsrc_prog.3.08

TurboACSource Programming : TurboACSource Connections : Connecting to the DUT Through TurboACSource Port

Connecting to the DUT Through TurboACSource Port

Each TurboACSource has four ports per the positive, negative, and reference pins as shown in the figure.



TurboACSource Multiplexed Channels

A TurboACSource can connect to any of the four Ports, but only one Port at the same time. If any pins are connected to a Port, then none of the other Ports can be used at that time.

Entries in the channel map and connections in VBT can differ depending on what connections to the DUT are required. Basic TurboACSource connections to the DUT are made as described in the following topics:

- [Connecting to the Channel Map Specified DIB Site](#)
- [Connecting to a Selected Port](#)

- [Connecting Individual Pins to a Port](#)

For connections made using the debug displays, see [Using the TurboACSource Debug Display](#).

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Connecting to the Channel Map Specified DIB Site

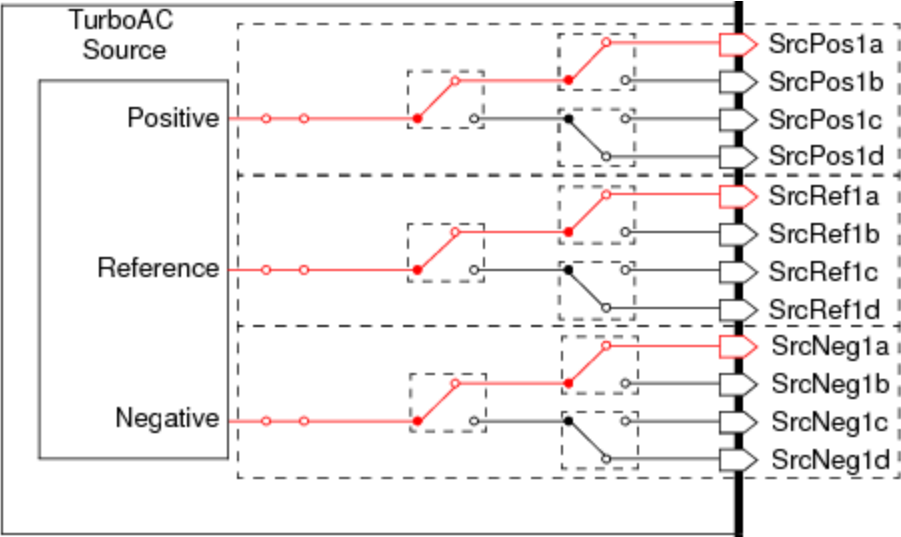
You can connect all three pins for a TurboACSource based on a single channel map entry to the DIB location specified in the Site column. The following channel map example shows a single pin entry for channel 1. The connection specified in the channel map is the positive source pin to Port A:

Pin Name	Package	Type	Site	Comment
MyPin		TurboACSrcPos	18.srcpos1a	

`tlTurboACSourceFromInst` determines that all pins for the channel are connected.

`tlTurboACSourceToDUT` determines that the Port specified in the channel map is connected.

The figure shows what connections are made to the DUT.



Connecting to Channel Map Specified Port

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TurboACSource Programming : TurboACSource Connections : Connecting to a Selected Port

Connecting to a Selected Port

You can connect all three pins for a TurboACSource to a port other than the one specified in the [channel map](#) Site column. The following channel map example shows a single pin entry for channel 1. The connection specified in the channel map is the positive source pin to Port A, but we will not connect to that pin:

Channel Map for Basic Connections

Pin Name	Package	Type	Site	Comment
MyPin		TurboACSrcPos	18.srcpos1a	

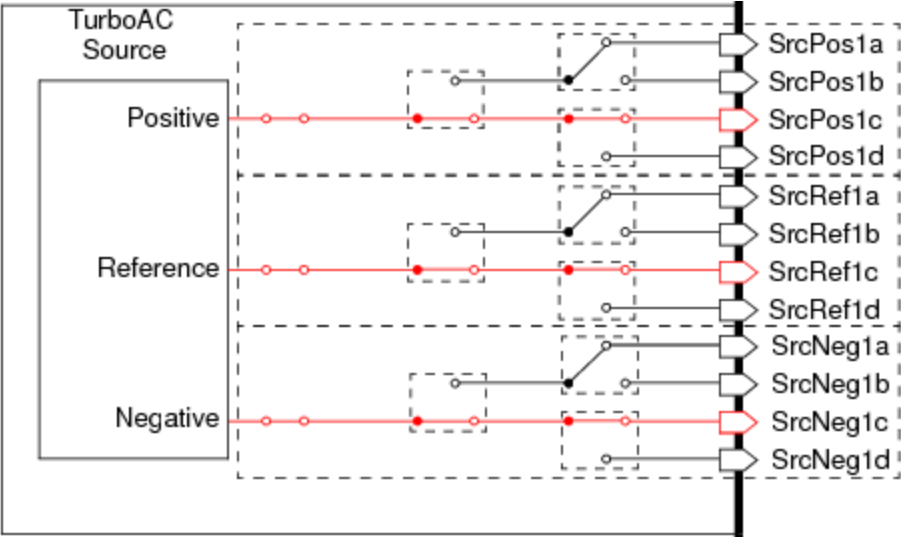
The following VBT connects to the DUT using a selected Port rather than the Port specified in the channel map:

```
TheHdw.TurboACSource.Pins("MyPin").Connect tlTurboACSourceFromInst _  
    , tlTurboACSourceToPortC
```

[tlTurboACSourceFromInst](#) determines that all pins for the channel are connected.

[tlTurboACSourceToPortC](#) determines that the pins are connected to port C although port A is specified in the in the channel map.

The figure shows what connections are made to the DUT.



Connecting to a Specified Port

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TurboACSource Programming : TurboACSource Connections : Connecting Individual Pins to a Port

Connecting Individual Pins to a Port

For certain tests, you may not want to connect all three TurboACSource pins to the DUT. To do this, each pin requires a separate entry in the channel map as shown in the following example:

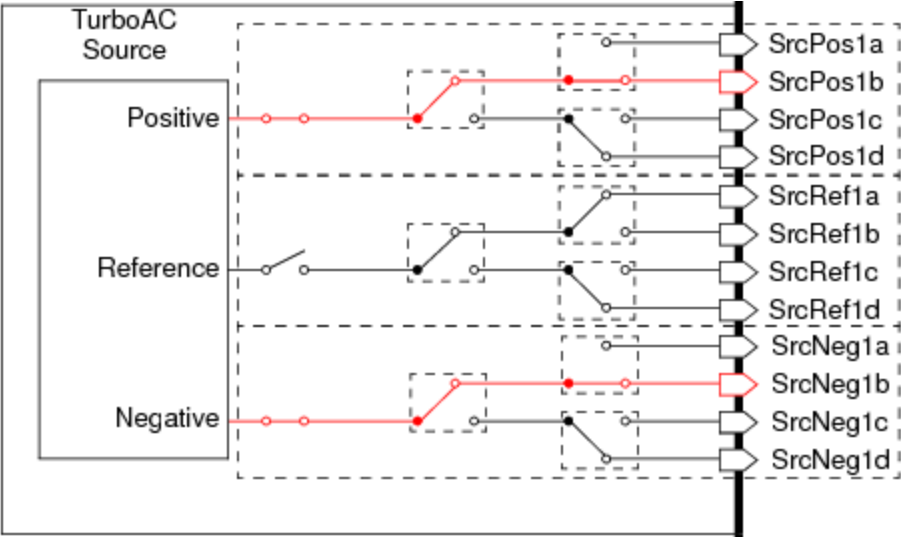
Channel Map for Basic Connections

Pin Name	Package	Type	Site	Comment
PosSrcPin		TurboACSrcPos	18.srcpos1a	
NegSrcPin		TurboACSrcNeg	18.srcneg1a	
RefSrcPin		TurboACSrcRef	18.srcref1a	

The following VBT syntax connects the positive and negative pin to the DUT but leaves the reference disconnected:

```
' Connect the positive pin to Port B
TheHdw.TurboACSource.Pins("MyPosPin").Connect tlTurboACSourceFromPOS _
    , tlTurboACSourceToPortA
' Connect the negative pin to Port B
TheHdw.TurboACSource.Pins("MyNegPin").Connect tlTurboACSourceFromNEG _
    , tlTurboACSourceToPortA
```

The figure shows what connections are made to the DUT.



Port Connections with Reference Disconnected

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TurboACSource Programming : TurboACSource Connections : BBAC Compatible Connections

BBAC Compatible Connections

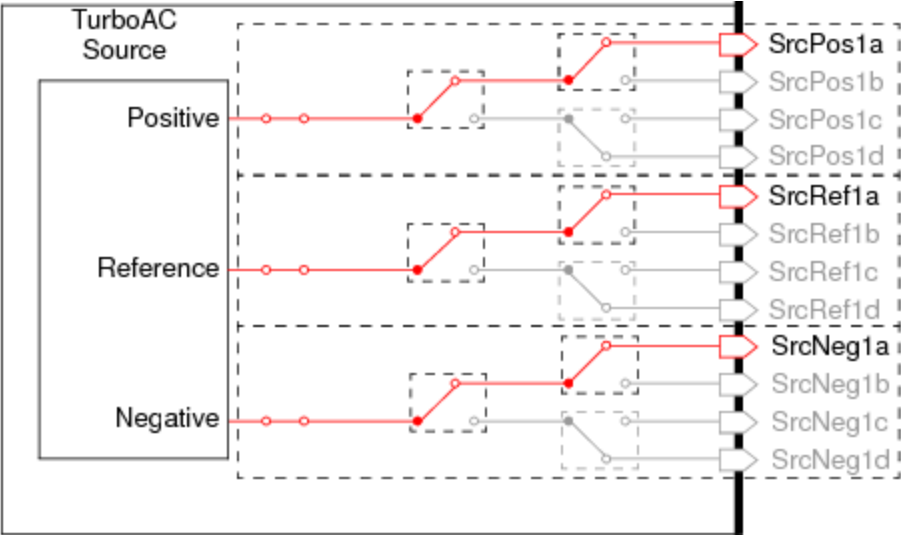
You can use the TurboACSource in a BBACSource compatible mode. When used as a BBACSource, only the A port can be used. The following displays a channel map configured for BBAC compatible mode:

Channel Map Example of BBAC Compatible Mode				
Pin Name	Package	Type	Site	Comment
MyPosPin		BBACSrcPos	18.srcpos1	
MyNegPin		BBACSrcNeg	18.srcneg1	
MyRefPin		BBACSrcRef	18.srcref1	

When using the TurboACSource in BBAC compatible mode, you program the instrument as if it were a BBAC instrument. The Pogo site location used is that of the Turbo Frame board, but the BBAC type, VBT syntax, and debug displays are used to program the instrument. The following syntax connects the TurboACSource in BBAC compatible mode for the above channel map:

```
TheHdw.BBACSource.Pins("MyPosPin").Connect tIBBACSourceFromInst _  
    , tIBBACSourceToDUT
```

The following figure shows how the ports are used in BBAC compatible mode.



TurboACSource Connections in BBAC Compatible Mode

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[TurboACSource Programming : TurboACSource Connections](#) : Making a Differential Connection to Capture Cancellation Circuitry

[Making a Differential Connection to Capture Cancellation Circuitry](#)

You can connect the TurboACSource to the cancellation circuitry of the TurboACCapture instrument. See [Signal Cancellation](#) for a description of this feature.

During a differential signal cancellation, the TurboACSource connects to the TurboACCapture cancellation circuitry and the TurboACCapture connects differentially to the DUT.

The following VBT syntax is used to make these required connections:

' Connect the TurboACSource to the TurboACCapture

```
TheHdw.TurboACSource.Pins("SrcPosPin").Connect tITurboACSourceFromInst _
    , tITurboACSourceToCaptureCancellation
```

' Enable the TurboACCapture cancellation circuitry

```
TheHdw.TurboACCapture.Pins("CapPosPin").SignalCancellationEnabled = True
```

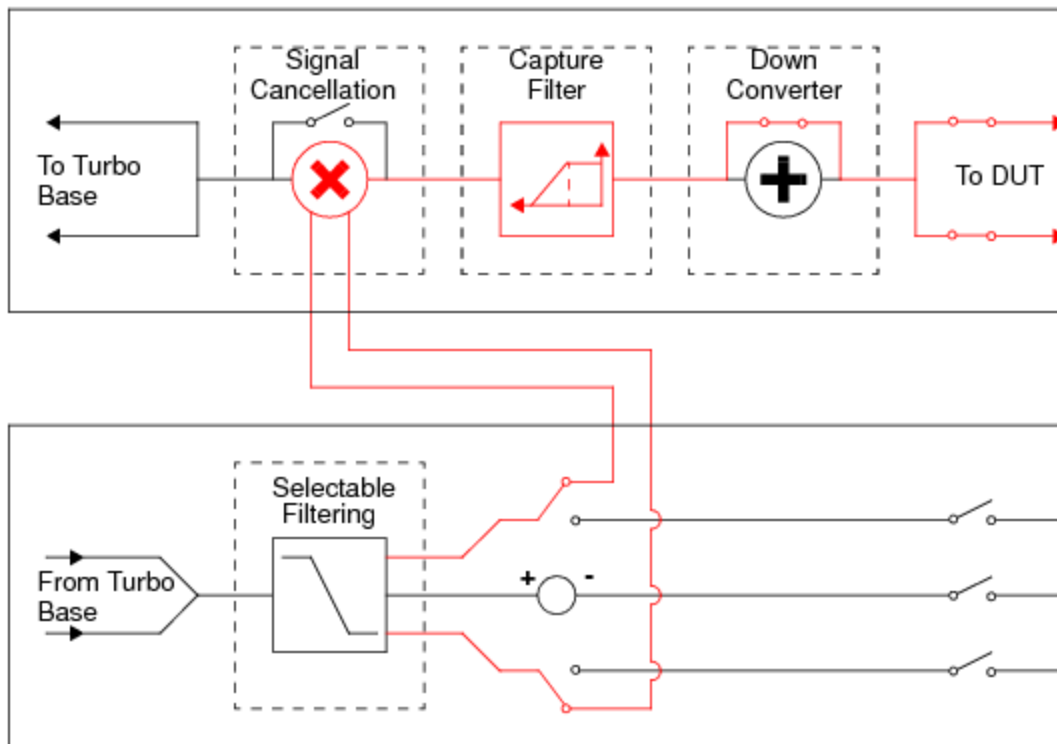
' Connect the TurboACSource to the DUT allowing the signal to be input

```
TheHdw.TurboACCapture.Pins("CapPosPin").Connect tITurboACCaptureFromInst,
tITurboACCaptureToDUT
```

Note:

The TurboACSource must connect differentially to the cancellation circuitry. You cannot connect only the positive pin or the negative pin. However, you can cancel a single-ended signal from the DUT.

The figure shows the connections between the TurboACSource and TurboACCapture in a differential signal cancellation setup. Note that the TurboACCapture has both the positive and negative pin connected.



TurboACSource Differential Connection to Signal Cancellation

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TurboACSource Programming : TurboACSource Connections : Making a Single-Ended Connection to Capture Cancellation Circuitry

Making a Single-Ended Connection to Capture Cancellation Circuitry

You can connect the TurboACSource to the cancellation circuitry of the TurboACCapture instrument. See [Signal Cancellation](#) for a description of this feature.

During a single-ended signal cancellation, the TurboACSource connects to the TurboACCapture cancellation circuitry differentially, and the TurboACCapture connects the required pin to the DUT and the other pin to the CC Common.

The following VBT syntax makes these required connections:

' Connect the TurboACSource to the TurboACCapture

```
TheHdw.TurboACSource.Pins("SrcPosPin").Connect tITurboACSourceFromInst _
    , tITurboACSourceToCaptureCancellation
```

' Enable the TurboACCapture cancellation circuitry

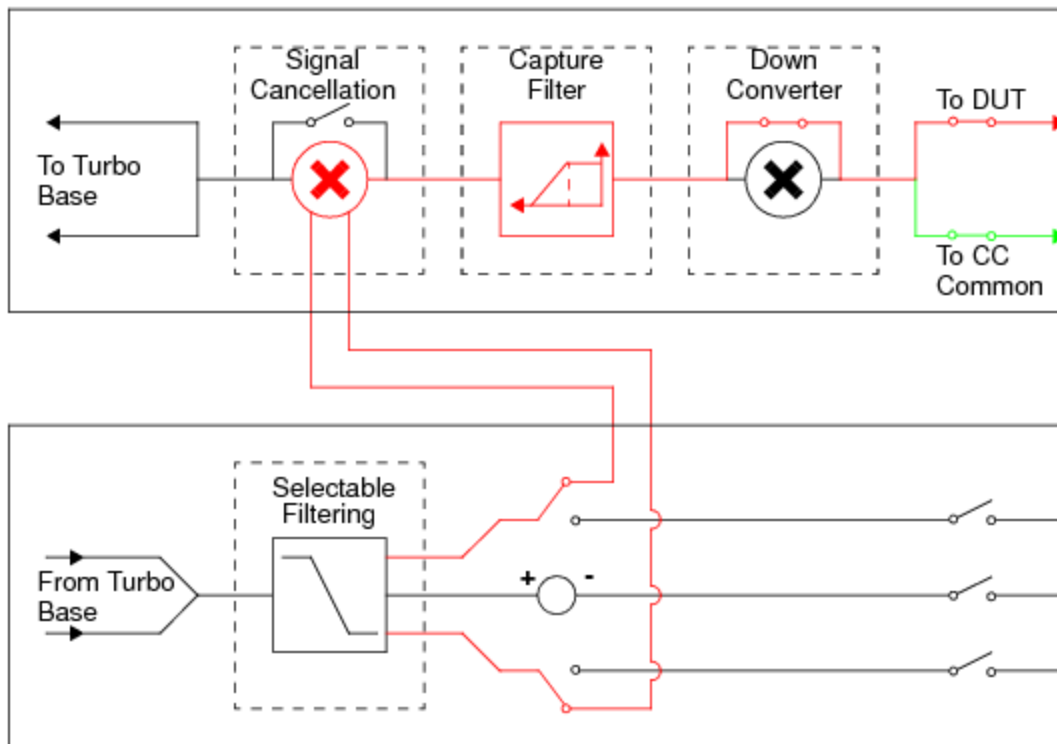
```
TheHdw.TurboACCapture.Pins("CapPosPin").SignalCancellationEnabled = True
```

' Connect the TurboACSource to the DUT allowing the signal to be input

```
TheHdw.TurboACCapture.Pins("CapPosPin").Connect tITurboACCaptureFromPOS,
tITurboACCaptureToDUT
```

☑ **Note:**

The TurboACSource must connect differentially to the cancellation circuitry. You cannot connect only the positive pin or the negative pin. However, you can cancel a single-ended signal from the DUT.



TurboACSource Connected to Signal Cancellation

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TurboACSource Programming : TurboACSource Connections : Connecting a DUT Pin to CC Common

Connecting a DUT Pin to CC Common

You can connect a DUT pin through the positive, negative, or reference pin to the CC Common. Use a channel map and syntax similar to the following:

Channel Map for Basic Connections				
Pin Name	Package	Type	Site	Comment
MyPin		TurboACSrcNeg	18.srcneg1c	

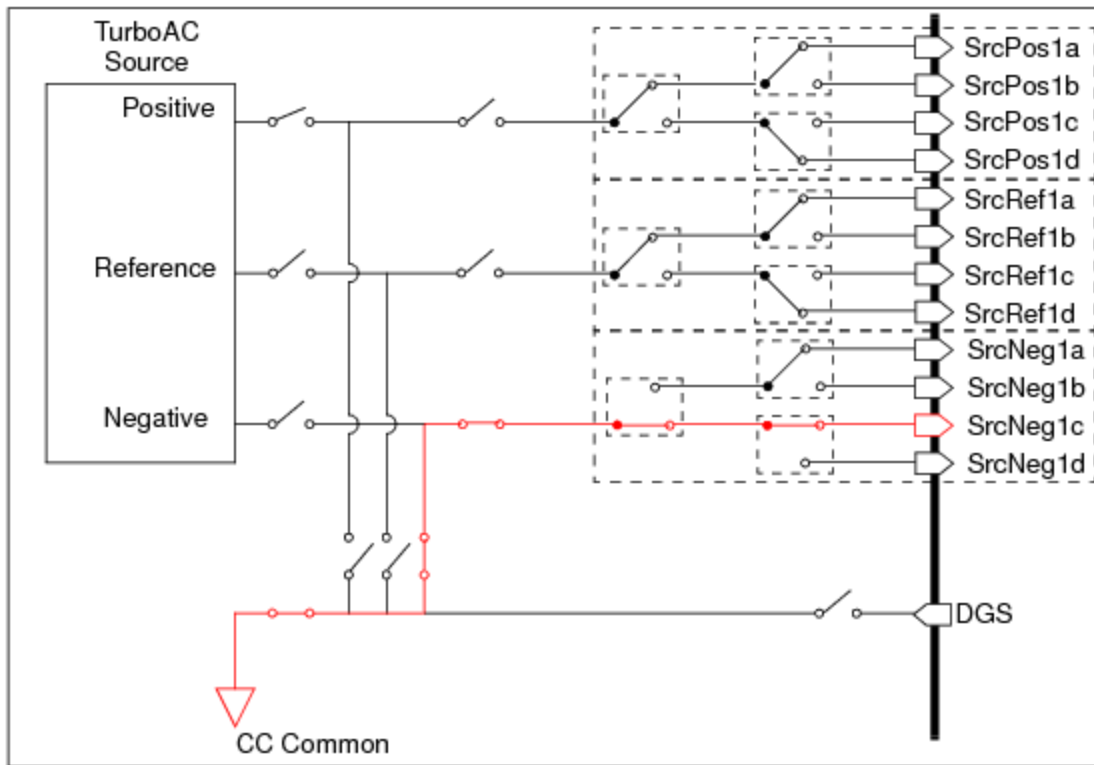
This VBT syntax makes connections between the DUT pin connected to pin SrcNeg1c and CC Common:

TheHdw.RefGround.Pins("MyPin").Connect tlCCGroundTypeCCC

tlCCGroundTypeCCC determines that the DUT pin is connected for the channel.

The port connected is determined by the channel map DIB Site Location Site.

The following figure shows what connections would be made.



Connection to CC Common

The TurboACSource does not support the following syntax:

`TheHdw.RefGround.Pins(SrcPin).Connect tlCCGroundTypeDGS`

The DUT cannot connect to the instrument DGS.

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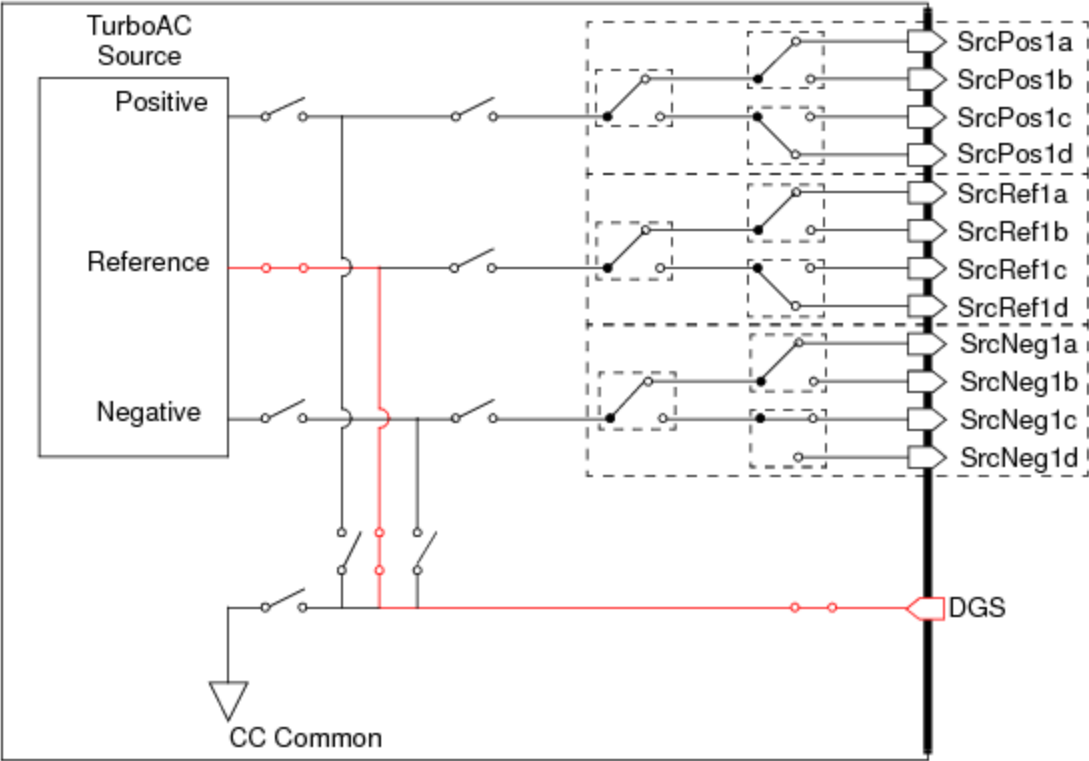
TurboACSource Programming : TurboACSource Connections : Connecting to DGS

Connecting to DGS

TurboACSource reference pins can connect to the instrument DGS. You can connect all pins or each individually. The table shows an example of a channel map:

Channel Map for DGS Connections				
Pin Name	Package	Type	Site	Comment
MyPosPin		TurboACSrcPos	18.srcpos1a	
MyNegPin		TurboACSrcNeg	18.srcneg1a	
MyRefPin		TurboACSrcRef	18.srcref1a	

The VBT connects the reference pin to the channel DGS:
TheHdw.TurboACSource.Pins("MyRefPin").Connect tlTurboACSourceFromREF,
tlTurboACSourceToDGS
The figure shows the connection made:



Connection to DGS

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TurboACSource Programming : TurboACSource Connections : Connecting the Loopback Circuitry

Connecting the Loopback Circuitry

The TurboAC instrument supports internal analog loopback between the source and capture. For more information, see Internal Loopback Circuitry.

A standard loopback connection is made between source1 and capture1 or source2 and capture2. Make the loopback connection using the channel map specified DIB location. The Port that is specified does not matter for this connection, although the channel number specified does.

Channel Map for Basic Connections

Pin Name	Package	Type	Site	Comment
MySrcPin		TurboACSrcNeg	18.srcneg1c	
MyCapPin		TurboACCapNeg	18.capneg1c	

This VBT syntax connects source1 and capture1 in loopback:

' Connect both the positive and negative capture pins

```
TheHdw.TurboACCapture.Pins("MyCapPin").Connect tlTurboACCaptureFromInst, _  
    tlTurboACCaptureToLoopBack
```

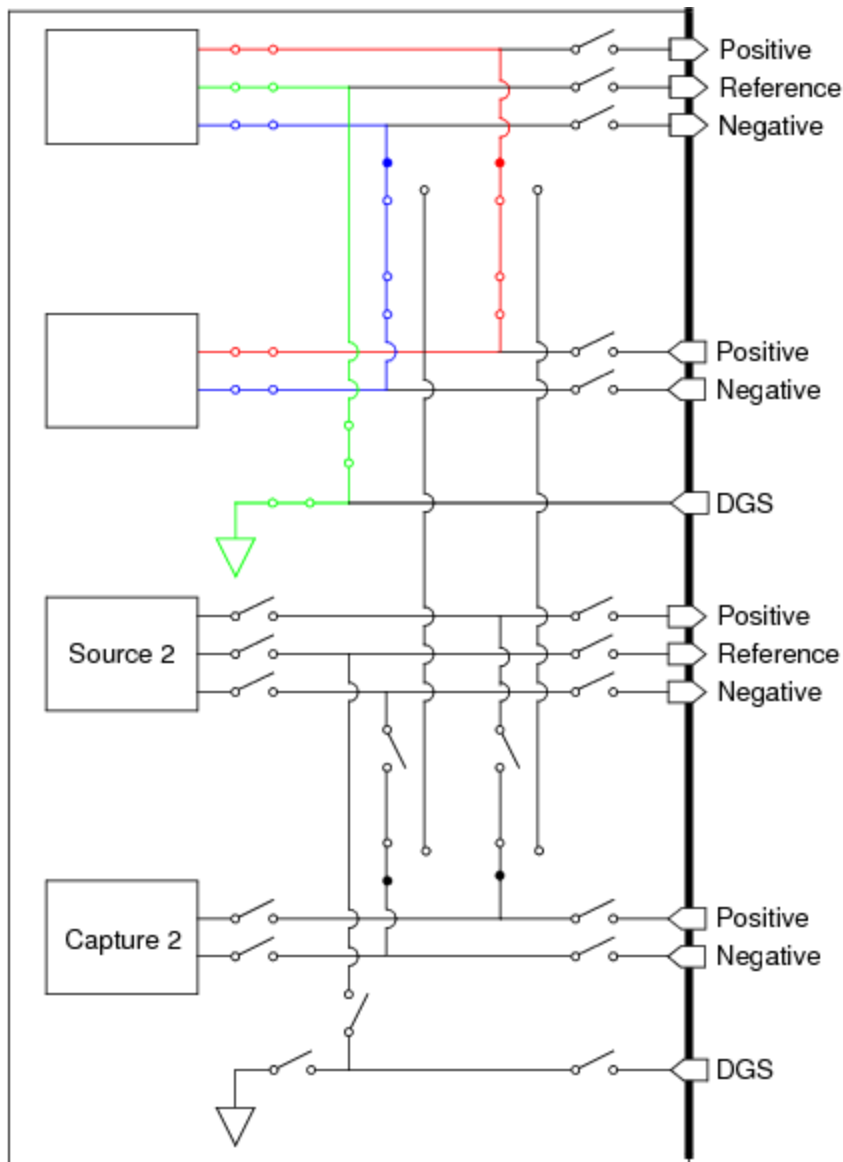
' Connect the positive and negative source pins, reference is not connected

```
TheHdw.TurboACSource.Pins("MySrcPin").Connect tlTurboACSourceFromNEG Or _  
    tlTurboACSourceFromPOS, tlTurboACSourceToLoopBack
```

' Connect reference to CC Common

```
TheHdw.TurboACSource.Pins("MySrcPin").Connect tlTurboACSourceFromREF, _  
    tlTurboACSourceToCCC
```

The figure shows what connections would be made.



Block Diagram of Standard Loopback Connection

The following code demonstrates disconnecting the TurboACSource and TurboACCapture from loopback:

' Disconnect both the positive and negative capture pins

```
TheHdw.TurboACCapture("MyCapPin").Disconnect tITurboACCaptureFromInst, _
    tITurboACCaptureToLoopBack
```

' Disconnect the positive and negative source pins

```
TheHdw.TurboACSource("MySrcPin").Disconnect tITurboACSourceFromNEG Or _
    tITurboACSourceFromPOS, tITurboACSourceToLoopBack
```

' Disconnect reference from CC Common

```
TheHdw.TurboACSource("MySrcPin").Disconnect tITurboACSourceFromREF, _
    tITurboACSourceToCCC
```

☑ **Note:**

Ensure that the TurboAC Source and Capture are disconnected from other signal paths (for example, from the DIB/DUT) to avoid external signal interference while using internal loopback.

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TurboACSource Programming : TurboACSource Connections : Connecting the Cross Channel Loopback Circuitry

Connecting the Cross Channel Loopback Circuitry

The TurboAC instrument supports internal analog loopback between the source and capture. For more information, see Internal Loopback Circuitry.

A cross channel loopback connection is made between source1 and capture2 or source2 and capture1.

The loopback connection is made using the channel map specified DIB location. The Port that is specified does not matter for this connection, although the channel number specified does.

Channel Map for Basic Connections

Pin Name	Package	Type	Site	Comment
MySrcPin		TurboACSrcNeg	18.srcneg1c	
MyCapPin		TurboACCapNeg	18.capneg2a	

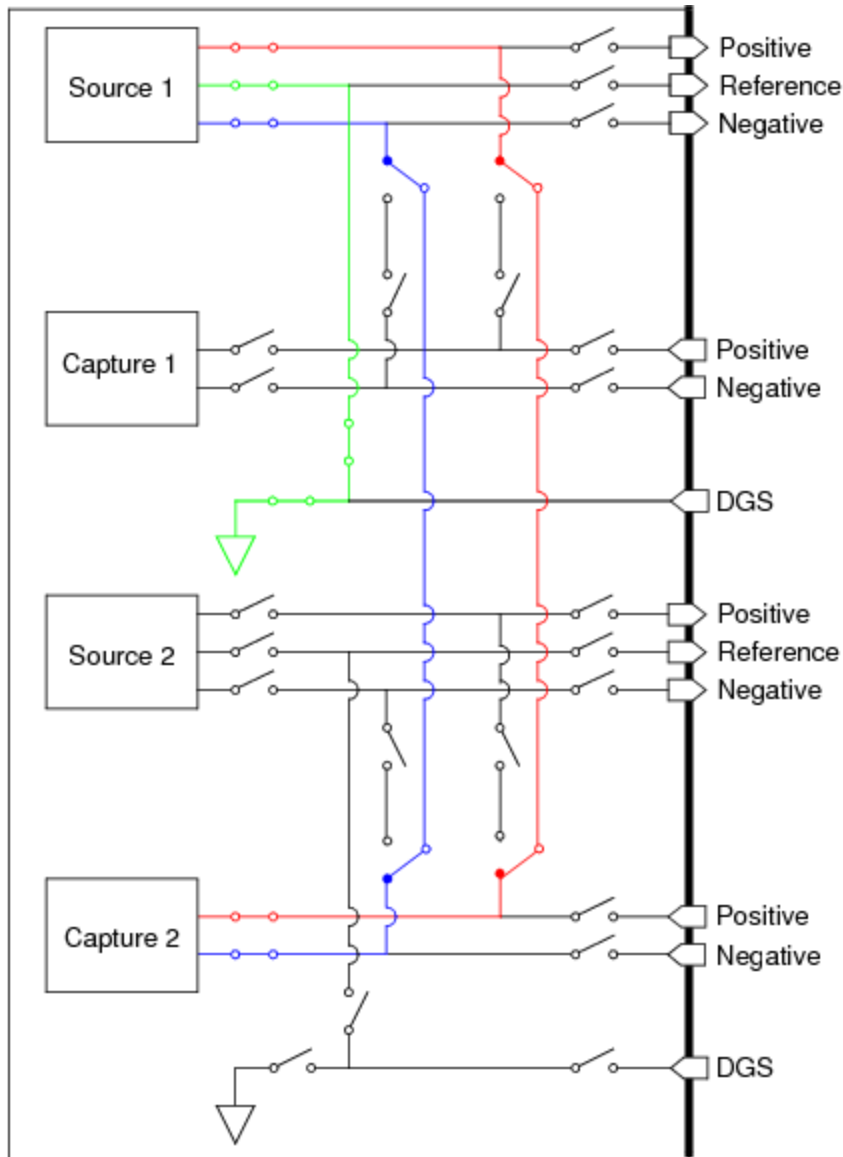
This VBT syntax connects source1 and capture2 in cross channel loopback:

```
' Connect both the positive and negative capture pins
TheHdw.TurboACCapture.Pins("MyCapPin").Connect tlTurboACCaptureFromInst, _
    tlTurboACCaptureToCrossChannelLoopBack

' Connect the positive and negative source pins, reference is not connected
TheHdw.TurboACSource.Pins("MySrcPin").Connect tlTurboACSourceFromNEG Or _
    tlTurboACSourceFromPOS, tlTurboACCaptureToCrossChannelLoopBack

' Connect reference to CC Common
TheHdw.TurboACSource.Pins("MySrcPin").Connect tlTurboACSourceFromREF, _
    tlTurboACSourceToCCC
```

The figure shows what connections would be made.



Block Diagram of Cross Channel Loopback Connection

The following code demonstrates disconnecting the TurboACSource and TurboACCapture from loopback:

' Disconnect both the positive and negative capture pins

```
TheHdw.TurboACCapture("MyCapPin").Disconnect tITurboACCaptureFromInst, _
    tITurboACCaptureToCrossChannelLoopBack
```

' Disconnect the positive and negative source pins

```
TheHdw.TurboACSource("MySrcPin").Disconnect tITurboACSourceFromNEG Or _
    tITurboACSourceFromPOS, tITurboACCaptureToCrossChannelLoopBack
```

' Disconnect reference from CC Common

```
TheHdw.TurboACSource("MySrcPin").Disconnect tITurboACSourceFromREF, _
    tITurboACSourceToCCC
```

☒ **Note:**

Ensure that the TurboAC Source and Capture are disconnected from other signal paths (for example, from the DIB/DUT) to avoid external signal interference while using internal loopback.

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TurboACSource Programming : TurboACSource Connections : Connecting to a PPMU

Connecting to a PPMU

You can connect each TurboACSource channel to a Pin Parametric Measurement Unit (PPMU) circuit for performing continuity tests.

Make connections when the TurboACSource is in ByPass mode using the following syntax:

TheHdw.PPMU.Pins(PinList).Connect

The PinList can be a pin group of TurboACSource pins, PPMU pins, or a combination of both.

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TurboACSource Programming : TurboACSource Connections : DIB Access Connections

DIB Access Connections

When a device pin needs multiple or different resources, you can use DIB access to minimize the number of DIB relays that are required and to create a simpler programming model. Once the channel map is set up to indicate the DIB connectivity, IG-XL addresses the correct resource based solely on the pin name. You need not create a virtual pin to reference the different resources.

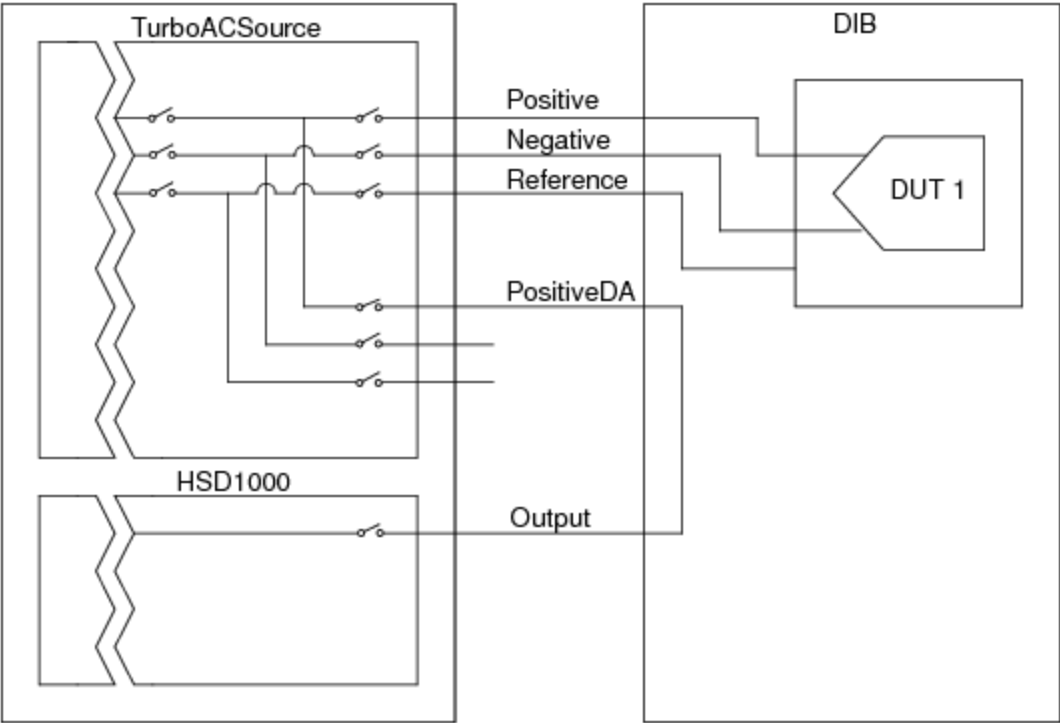
- | | |
|--|---|
| | <ul style="list-style-type: none">TurboACSource as a Primary Resource |
| | <ul style="list-style-type: none">TurboACSource as Secondary Resource |

☒ **Note:**

Only one DIB Access connection is shared across the four ports. You cannot use DIB access with TurboACSource filters or signal combination.

TurboACSource as a Primary Resource

When the TurboACSource is the primary instrument, this means that another instrument pin enters the TurboACSource DIB access pin and connects to the DUT through the TurboACSource DUT connection. The figure shows a basic block diagram of this connection.



TurboAC as Primary Instrument in a DIB Access Connection

When the TurboACSource is in BBAC compatible mode, DIB Access connections are programmed like BBAC DIB Access connections. The following shows a pin map, channel map, VBT syntax, and a diagram of the connections made for an HSD resource delivered as secondary resources to the device pin.

Only a single pin name is required in the pin map. The pin type used is that of the primary instrument although this is unnecessary. Remember that selecting a certain pin type can change the available pin levels selections.

Pin Map for BBAC Source as a Secondary Resource

Group Name	Pin Name	Type
	InPos	Analog

The same pin name is entered in the channel map twice. Once with the type of BBACSrcPos, and once with I/O for the digital pin. The :DA syntax indicates that the I/O connection will be made through DIB access.

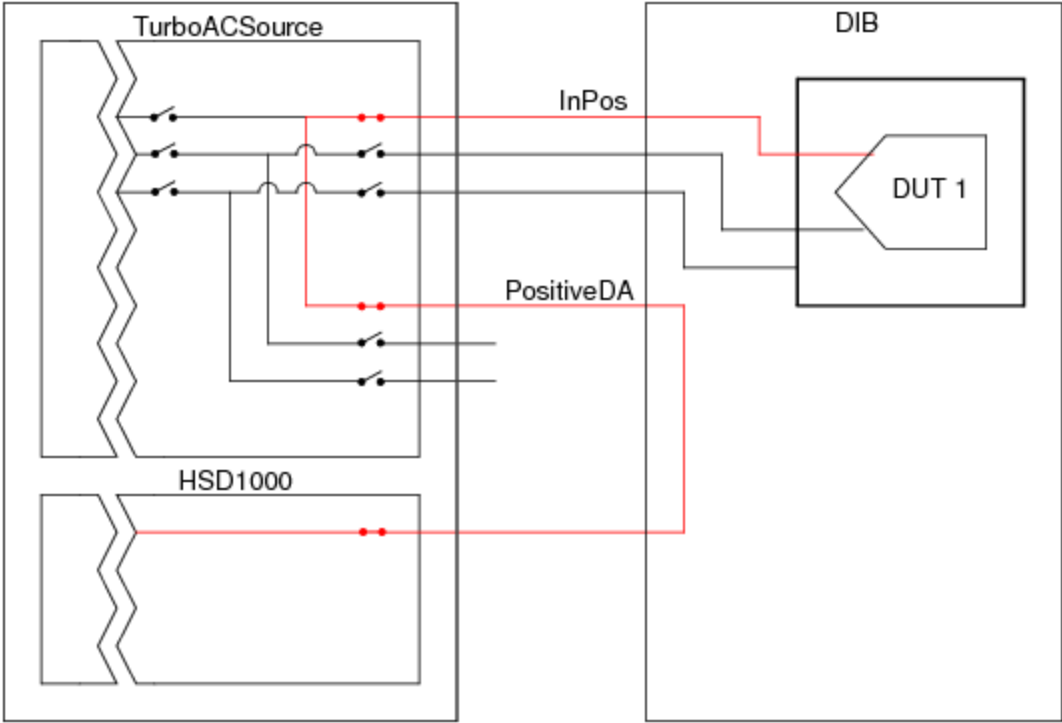
Channel Map for BBAC Source as a Primary Resource

Pin Name	Package Pin	Type	Site 0
InPos		BBACSrcPos	18.srcpos1
InPos:DA		I/O	1.ch1

The following VBT syntax connects to the Digital Pin to the DUT through the TurboACSource connection:

```
TheHdw.Digital.Pins("InPos").Connect
```

The diagram show the connection and signal flow that is created.



DIB Access Connection, TurboACSource as a Primary Resource

Note:

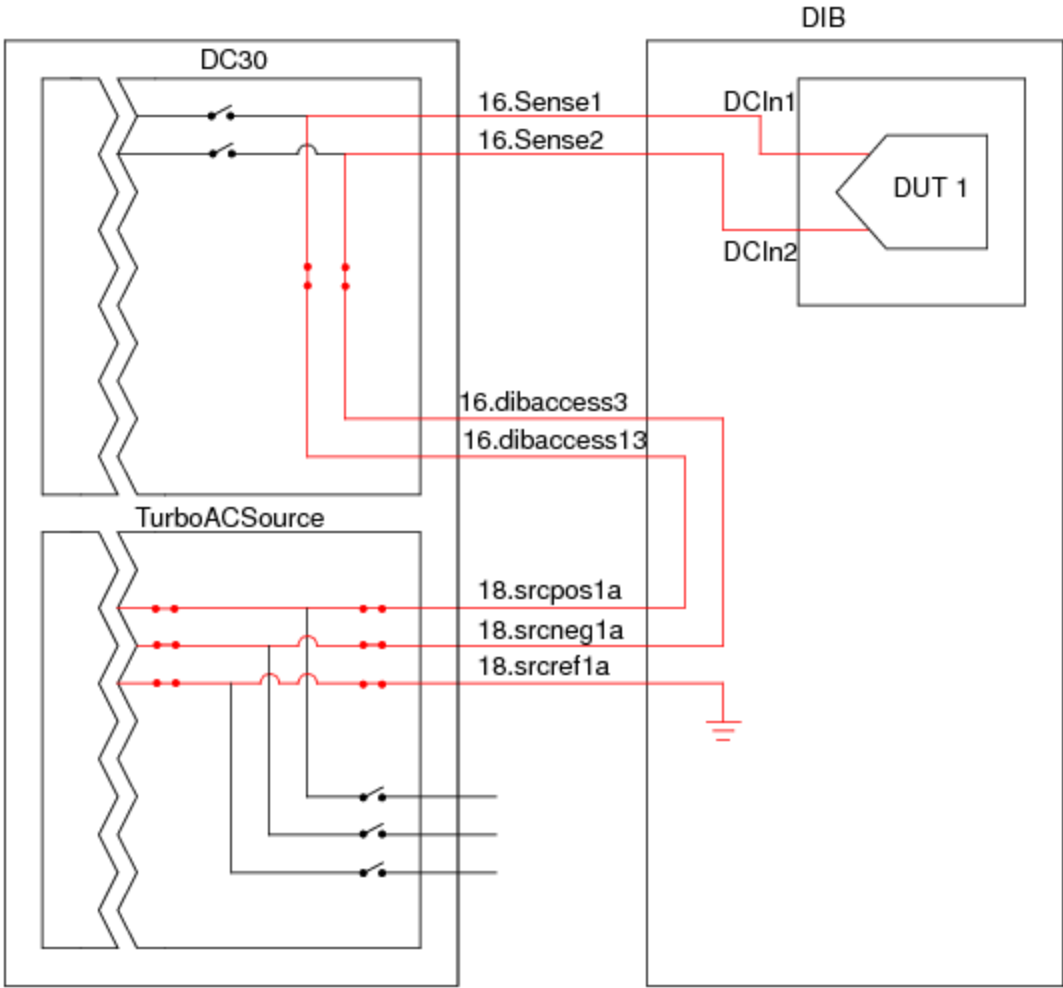
Proper DIB layout is required when making DIB access connections.

TurboACSource as Secondary Resource

The DIB access use model for the TurboACSource as a secondary resource for a device pin is identical to other UltraFLEX instruments. This figure shows a TurboACSource delivered as a secondary resource to the device pins using DIB access.

Channel Map for TurboACSource as a Secondary Resource

Pin Name	Package Pin	Type	Site 0
DC_IN1		DCVI	16.sense1
DC_IN1:DA		TurboACSrcPos	18.srcpos1a
DC_IN2		DCVI	16.sense2
DC_IN2:DA		TurboACSrcNeg	18.srcneg1a



TurboAC DIB Access, TurboACSource as a Secondary Resource

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[TurboACSource Programming](#) : Visual Basic for Test

Visual Basic for Test

You create TurboACSource programs using VBT syntax in the VBT editor.

For TurboACSource specific VBT programming, see [TurboACSource](#) in the VBT Syntax Reference.

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TurboACSource Programming : TurboACSource Reset Values

TurboACSource Reset Values

The table lists the local reset values for the TurboACSource. Pre-job reset, post-job reset, power-up reset, jobload reset, and user reset all result in the same state, unless otherwise noted.

TurboACSource Parameter Reset Values

TurboACSource Parameter	Reset Value
Common mode	0.0 V
Common mode enabled	True
Filter lowpass bypass	True
Filter lowpass value	1.25 MHz
Connection type	tITurboACSourceDifferential
Signal source	tITurboACSourceAWG

AWG Signal Source Parameter Reset Values

AWG Signal Source Parameter	Reset Value
Amplitude	1.0 V
Dither enabled	False
External trigger	The loaded command (if any) is removed.
Microcode control	tITurboACSourceMicrocodeControlByCPU

Sample rate	10 MHz
Voltage range	Auto (1.28 V)

TurboACSource Standard CW Parameter Reset Values

TurboAC Standard CW Source Parameter	Reset Value
Amplitude	0.0089V Peak
Recalibration limit amplitude	0.1dBm
Recalibration limit frequency	100Hz
Frequency	1.0 MHz
Load impedance	50ohms

TurboACSource Optional CW Parameter Reset Values

TurboAC Optional CW Source Parameter	Reset Value
Amplitude	0.0089V Peak
Recalibration limit amplitude	0.1dBm
Recalibration limit frequency	100Hz
Frequency	1.0MHz
Load impedance	50ohms

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TurboACSource Programming : TurboACSource Pattern Control

TurboACSource Pattern Control

When in AWG or direct capture mode you can control a TurboACSource instrument using pattern microcodes, allowing changes to instrument states and parameters during a program. You can obtain repeatable timing by programming an instrument using pattern microcodes and parameter sets (PSets).

You create a pattern independently of the test program using the PatternTool or a text editor. Patterns are composed of a series of numbered digital vectors. These digital vectors define the data for each channel listed in the pattern, as well the pattern microcode used for controlling pattern execution.

☒ **Note:**

To use pattern microcodes to control the TurboACSource on UltraFLEX, you must include a digital pin in the following places: the pin map, the channel map, and the pin list of the pattern module. The two forms of pattern control are:

Microcode	A microcode controls a single function.
PSet	A parameter set is a unique microcode that contains a set of values that describe a specific instrument setup. PSets allow a test program to change instrument setups within a pattern execution. Although a PSet is a hardware capability that changes the setup of an instrument during a pattern burst, you define it using VBT language and invoke it using PSet microcode.

The following topics are available:

- [Programming Patterns in a Text Editor](#)
- [TurboACSource Microcodes](#)
- [PSets](#)

[Pattern Compiler](#)

The [pattern compiler](#) compiles a [text .atp](#) file into a binary [.pat](#) file. You can edit this file using [PatternTool](#), and it is the format accepted by the pattern loader.

[Pattern Reverse Compiler](#)

The [pattern reverse compiler](#) compiles a [binary .pat](#) file into a [text .atp](#) file. You can edit this file using a text editor.

[PatternTool](#)

[PatternTool](#) allows you to read and edit binary pattern files. Using [PatternTool](#), you can display a compiled pattern file in easily readable, column formats. You can also directly edit a pattern file, instead of having to edit the ASCII source file and recompile it. You can then update the tester with the edited pattern file.

[PatternTool](#) is useful during debugging, specifically in debugging pattern files. [PatternTool](#) can also display [HRAM \(History RAM\)](#) data, as captured on the tester.

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[TurboACSource Programming : TurboACSource Pattern Control](#) : Programming Patterns in a Text Editor

Programming Patterns in a Text Editor

You can create and edit a pattern using a text editor. This form (.atp files) must be compiled into a .pat file before it can be loaded. The pattern contains microcode to set instrument parameters. Use the PSet microcode to program the instrument to the parameters specified in a PSet.

Creating a pattern in this form requires you to format the file in a specific way. A pattern requires that an instrument statement for each instrument is added and that each vector is written out in a particular way.

The following topics show how to create a pattern using a text editor:

- [Instruments Statement](#)
- [Vector Statements](#)

Instruments Statement

You can only use TurboACSource microcodes with pins that the instruments statement has specified as TurboACSource. For example:

```
instruments = {  
    DUT_AIN:TurboACSource;  
    DUT_AIN:BBACSource;  
}
```

For more information, see [Instruments Statement](#) in the Pattern Language section.

Vector Statements

If a vector contains any TurboACSource related microcode other than nop, that vector is placed in SRM.

The format for specifying a TurboACSource microcode on a particular vector is:

(pin-list : TurboACSource = microcode)

The following code shows the ASCII pattern microcode syntax for the TurboACSource and Capture instruments. The device analog input (DUT_AIN) and analog output (DUT_AOUT) are connected to a TurboACSource and a TurboACCapture respectively.

```
instruments = {
    DUT_AOUT:TurboACCapture;
    DUT_AIN:TurboACSource;
}
vector ( $tset, Digital1, Digital2)
{
    > tset1 0      1      ;
    (DUT_AOUT = Resync)
    (DUT_AIN = Resync)
    > - 1      1      ;
    (DUT_AOUT = Trig )
    (DUT_AIN = Start )
    > - 0      1      ; /* Trigger/Start the default signal */
    (DUT_AOUT = Trig Audio_1kHz)
    (DUT_AIN = Start Audio_1kHz)
    > - 1      1      ; /* Trigger/Start a named signal */
    (DUT_AOUT = Enable_Alarm)
    (DUT_AIN = Start1 )
    > - 0      1      ;
    (DUT_AOUT = Disable_Alarm)
    (DUT_AIN = StartE )
    > - 1      1      ;
    (DUT_AOUT = Enable_Inst_Cond)
    (DUT_AIN = Start1E )
    > - 0      -      ;
    (DUT_AOUT = Disable_Inst_Cond)
    (DUT_AIN = Stop)
    > - 1      -      ;
    (DUT_AIN = StopE)
    > - 0      -      ;
    (DUT_AOUT = PSet AudioPSet)
    (DUT_AIN = PSet AudioPSet)
```

```
> - 1 - ;  
> - - - ;  
}
```

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TurboACSource Programming : TurboACSource Pattern Control : TurboACSource Microcodes

TurboACSource Microcodes

Microcode commands are available for use inside a pattern. Patterns allow you to control the precise timing of when a signal is sourced. To establish repeatable timing, or to synchronize with other instruments (such as HSD) in the tester, use a pattern. If you want to start generating a signal but are not concerned with precise timing, however, then you do not need a pattern. The table lists the microcodes used to issue commands to a TurboACSource instrument from a pattern.

TurboACSource Pattern Microcodes

Microcode	Description
PSet <PSetName>	Executes the specified parameter set (PSet) which changes the instrument’s setup (voltage range, amplitude, and sample rate). This microcode requires a single argument, which is the name of the PSet to be applied.
Resync	Forces phase alignment of the TurboAC source instrument clock with the HSD T0 clock.
Start <SignalName>	The instrument immediately (on the next available logical analog clock) starts sourcing the named signal in SMEM and repeats it continuously until a Stop microcode is executed, the instrument is reset, or another signal is started. This microcode takes an optional argument, which is the name of the signal to be sourced. If the signal name is not specified, the default signal is used.

StartE <SignalName>	At the end of the currently sourcing signal's segment, the instrument sources the specified signal in SMEM and repeats it continuously until a Stop microcode is executed, the instrument is reset, or another signal is started. This microcode takes an optional argument, which is the name of the signal to be sourced. If the signal name is not specified, the default signal is used.
Start1 <SignalName>	The instrument immediately starts sourcing (on the next available logical analog clock) the specified signal in SMEM and sources it exactly once. This microcode takes an optional argument, which is the name of the signal to be sourced. If the signal name is not specified, the default signal is used.
Start1E <SignalName>	At the end of the currently sourcing signal's segment, the instrument sources the specified signal in SMEM and sources it exactly once. This microcode takes an optional argument, which is the name of the signal to be sourced. If the signal name is not specified, the default signal is used.
Stop	The instrument stops sourcing the currently sourced signal immediately. Stop does not affect a signal that is running once.
StopE	The instrument stops sourcing the currently sourced signal at the end of the signal.

Text Editor Example

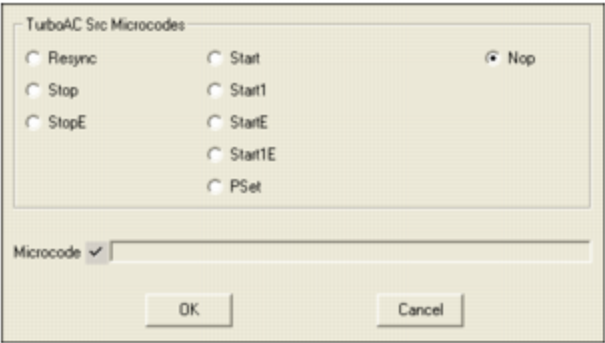
The following example displays how the TurboACSource microcode would appear in a pattern file created using a text editor:

```
(pin1:TurboACSource = StopE)    > -    ;
```

PatternTool GUI Example

The PatternTool provides columns for the TurboACSource pins that are listed in the instrument statement of the *.atp file.

The comma-separated microcode equivalent can be conveniently produced by selecting the check boxes as in this figure.



TurboACSource Microcode Display

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TurboACSource Programming : TurboACSource Pattern Control : PSets

PSets

PSets have programmable parameters to control the state of an instrument for testing. You define a PSet in the VBT code and invoke it using VBT or the PSet microcode from a pattern.

PSets allow you to specify several aspects of an instrument’s state at once.

Called From Text File

A PSet called from a text file would use the following format:

(MyPin:TurboACSource = PSet PSet1)

Called From PatternTool

To call a PSet in PatternTool the setup would look as follows:

TSet	Command	Label	Vector/Cy	srcipos (TurboACSource)	Comment
tset1			0		
tset1			1	PSet PSet1	
tset1			2		
tset1			3		

PSet Called Using PatternTool

PSets are programmed similarly in PatternTool or using a text editor.

Calling From VBT

A PSet applied with VBT language would use the following format:

TheHdw.TurboACSource.Pins("MyPin").PSets.Item("MyPSet").Apply

PSet Example

A PSet is defined in VBT and can look like the following example:

' Create a PSet and add it to the PSet collection

TheHdw.TurboACSource.Pins("MyPin").AWG.PSets.Add "MyPSet"

' Program the hardware parameters to the desired values

With TheHdw.TurboACSource.Pins("MyPin").AWG.PSets.Item("MyPset")

.SampleRate = 10000

```
.Amplitude _  
.PeakToPeak = 8  
End With  
' Load the PSet into PSet memory  
TheHdw.TurboACSource.Pins("MyPin").AWG.PSets.Item("MyPSet").Apply
```

TurboACSource PSets

Property/Method	Description
Amplitude	An interface to program the amplitude value of the PSet.
Amplitude.Peak	An interface to program the amplitude in Peak.
Amplitude.PeakToPeak	An interface to program the amplitude in peak to peak.
Amplitude.RMS	An interface to program the amplitude in RMS.
SampleRate	An interface to set the sample rate parameter of a PSet.
VoltageRange	An interface to program the voltage range of the specified PSet. An error appears if the specified signal does not exist in the collection.

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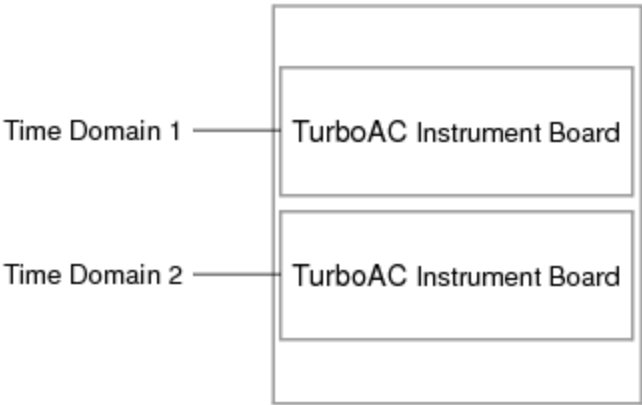
TurboACSource Programming : Using TurboAC in Concurrent Testing

Using TurboAC in Concurrent Testing

To support concurrent testing, you can assign each TurboAC instrument to a unique time domain. Concurrent features for TurboAC include:

- Each TurboAC board can be configured for one time domain
- Each TurboAC instrument can be controlled by a unique time domain using Sync-Link. Features include:
 - Independent access to Sync-Link
 - Independent sample clock programming, one sample rate per channel
 - Independent clock synchronization with other TurboAC instruments or with digital clocks
 - Independent control by digital pattern
 - Independent capture memory and triggers
- Multiple instruments can be grouped into a common time domain
- TurboAC is restricted to using the Global flow domain and synchronous pattern starts

The following figure shows a configuration with each TurboAC instrument assigned to its own time domain.



TurboAC Concurrent Testing Configuration Example, 2 Time Domains
For more information, see [About Concurrent Test](#).

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TurboACSource Theory of Operation

TurboACSource Theory of Operation

The TurboACSource instrument provides AC waveform source capabilities for testing devices such as precision A/D and D/A converters, high speed A/D and D/A converters, and high performance SOC's such as Set Top Box devices.

Typical tests performed using the TurboAC instruments include gain accuracy, frequency response, distortion, and signal-to-noise ratio.

The TurboACSource instrument can generate high-accuracy, low-noise and distortion signals with frequency components of up to 150MHz.

The following topics are available:

- TurboACSource Overview
- AWG Source
- Standard CW Source
- Optional CW Source
- DIB Source
- Signal Combination

The TurboACSource is a dual source per each TurboAC instrument board. Each board also includes two TurboACCapture instruments. For more information, see About the TurboACCapture.

For information on TurboACSource instrument performance specifications and standards, see [TurboAC Specifications](#).

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TurboACSource Theory of Operation : TurboACSource Overview

TurboACSource Overview

The TurboACSource instrument can generate high accuracy, low noise, and distortion signals with frequency components of up to 150MHz. The instrument features a programmable filter set for optimizing signal quality, as well as a combination circuit for creating dual tone signals from two source inputs. The TurboACSource output can be switched to one of four differential ports (eight single-ended) using a multiplexed relay tree.

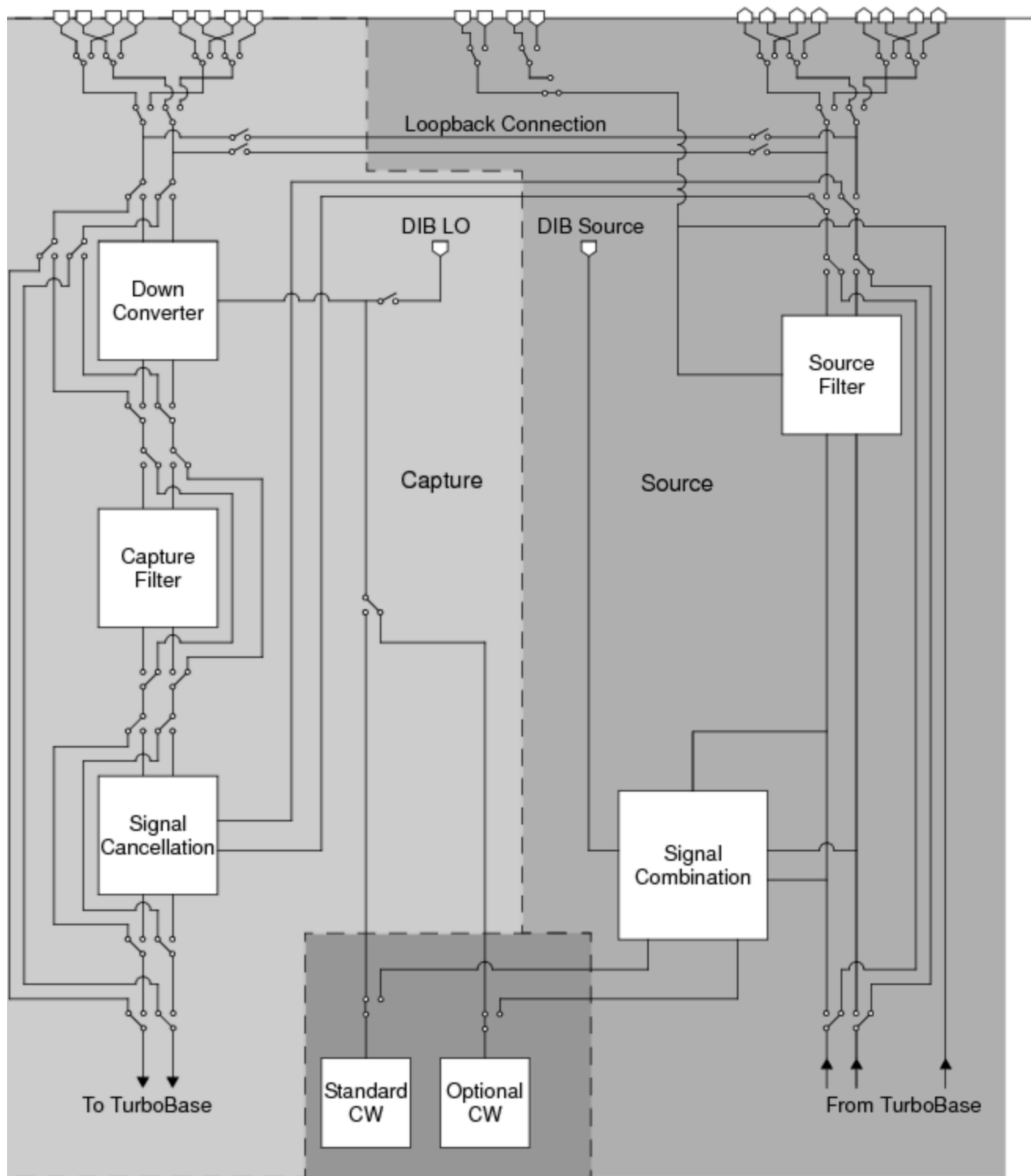
You program the TurboACSource using the VBT programming syntax, and you can also control the AWG using pattern microcodes and PSets. The TurboACSource includes an interactive debug display.

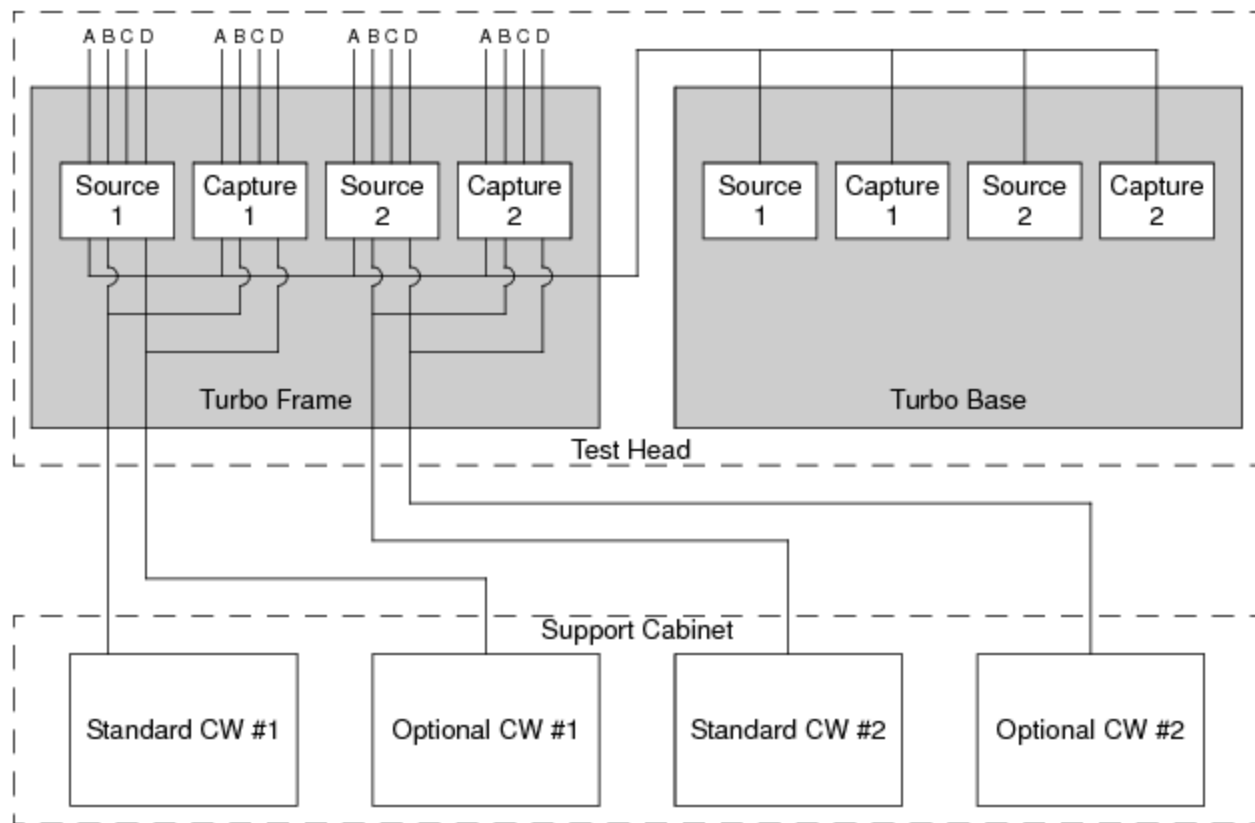
The TurboACSource offers the following modes of operation:

AWG	The TurboACSource performs as an arbitrary waveform generator (AWG) capable of generating signals up to 15MHz.
Standard CW	The TurboACSource performs as a continuous wave (CW) synthesizer capable of producing signals from 1MHz to 150MHz, although it is programmable to 310MHz.
Optional CW	The TurboACSource can deliver a signal from an alternative CW synthesizer through its auxiliary input port. The optional CW can be a Teradyne provided CW synthesizer that is integrated in IG-XL or a user supplied optional CW.
DIB	The TurboACSource lends its filters and combiner circuit to other test system instruments through its DIB input.

Signal Combination The TurboACSource's signal combination circuit creates dual tone signals from two individual sources (AWG, CW, Optional CW, or DIB).

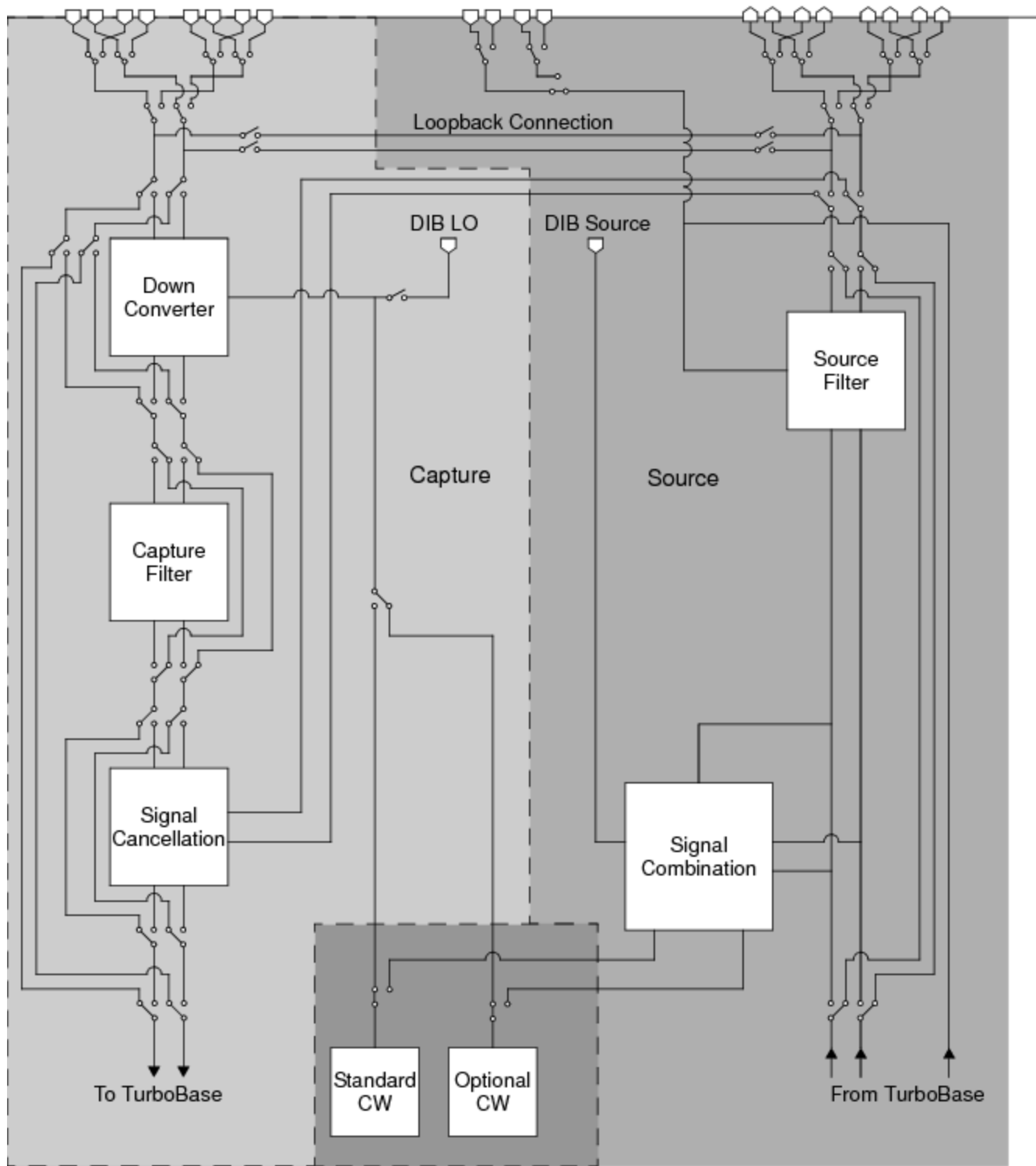
The TurboACSource hardware set consists of a Turbo Frame Board, a Turbo Base Board and optional devices from the support cabinet. The following is a functional diagram.





System Level Diagram of TurboAC Hardware

The following is a block diagram of the TurboAC Frame Board.



Single Channel on TurboAC Frame Board

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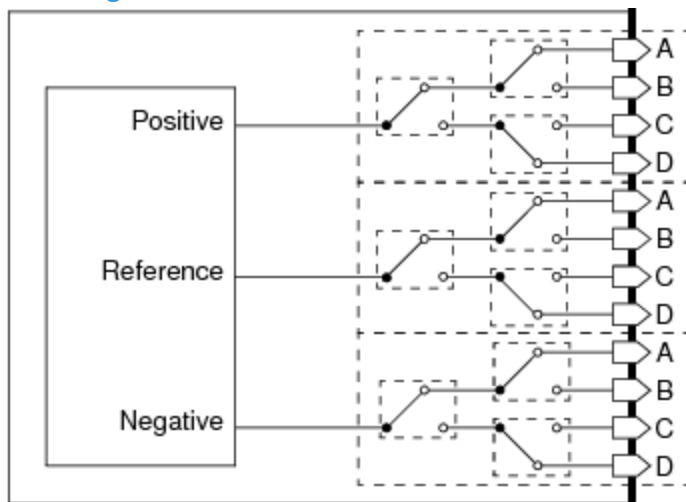
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[TurboACSource Theory of Operation](#) : [TurboACSource Overview](#) : TurboACSource Port

TurboACSource Port

Each TurboACSource connects to the DIB through a Port. A Port consists of multiplexed channels allowing each channel to connect to one of four outputs. The figure shows details of the Port.



TurboACSource Port Connections

When a connection is made to the DUT, all pins of a channel must connect to the same Port output. For example, if the positive pin is connected to the A Port, then the negative and reference pin must also be connected to the A Port.

If an illegal connection is attempted, an error is generated.

For more information, see [TurboACSource Connections](#).



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[TurboACSource Theory of Operation](#) : [TurboACSource Overview](#) : Internal Loopback Circuitry

Internal Loopback Circuitry

The TurboAC instrument supports internal analog loopback between the source and capture (Source 1 and Capture 1, Source 2 and Capture 2). This feature can be used to verify instrument performance and setup conditions without requiring a DIB or external instrumentation. Each source and capture instrument has an analog loopback relay path devoted to this function. You can source and capture a signal within the same instrument board, which eliminates influences outside of the instruments themselves.

For more information, see [Connecting the Loopback Circuitry](#).

Internal Cross Channel Loopback

Loopback can also be made by crossing to the opposite channel number (Source 1 and Capture 2, Source 2 and Capture 1). This feature is useful when either the source or capture in a standard loopback pairing is unavailable for use.

One use would be sourcing from the CW on source #1 and capturing with down conversion on capture #2, which cannot be done in channel loopback due to the single standard CW restriction. For more information, see [Connecting the Loopback Circuitry](#).

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[TurboACSource Theory of Operation](#) : [TurboACSource Overview](#) : [Capture Cancellation](#)

Capture Cancellation

The capture cancellation circuit eliminates a tone from a captured signal by using an output signal from the TurboACSource. The cancellation tone is added to the signal under test and is equal in magnitude and 180 degrees out of phase with the tone to be removed. The cancellation tone can come from any of the TurboACSource signal sources.

The ability to remove a tone depends on the instrument’s ability to create a signal of the same magnitude and opposite phase of the tone to be removed. The AWG is the easiest signal source to do this with.

One use of this feature is in intermodulation distortion testing. With one tone of a dual tone signal removed, any captured IMD products would be from the DUT and not the capture circuit. Another use is in testing harmonic distortion. Removing the fundamental tone allows the capture instrument to capture the device harmonics without generating any of its own.

For more information, see:

- | | |
|---|--|
| • | Making a Differential Connection to Capture Cancellation Circuitry |
|---|--|
- | | |
|---|-------------------------------------|
| • | Signal Cancellation |
|---|-------------------------------------|

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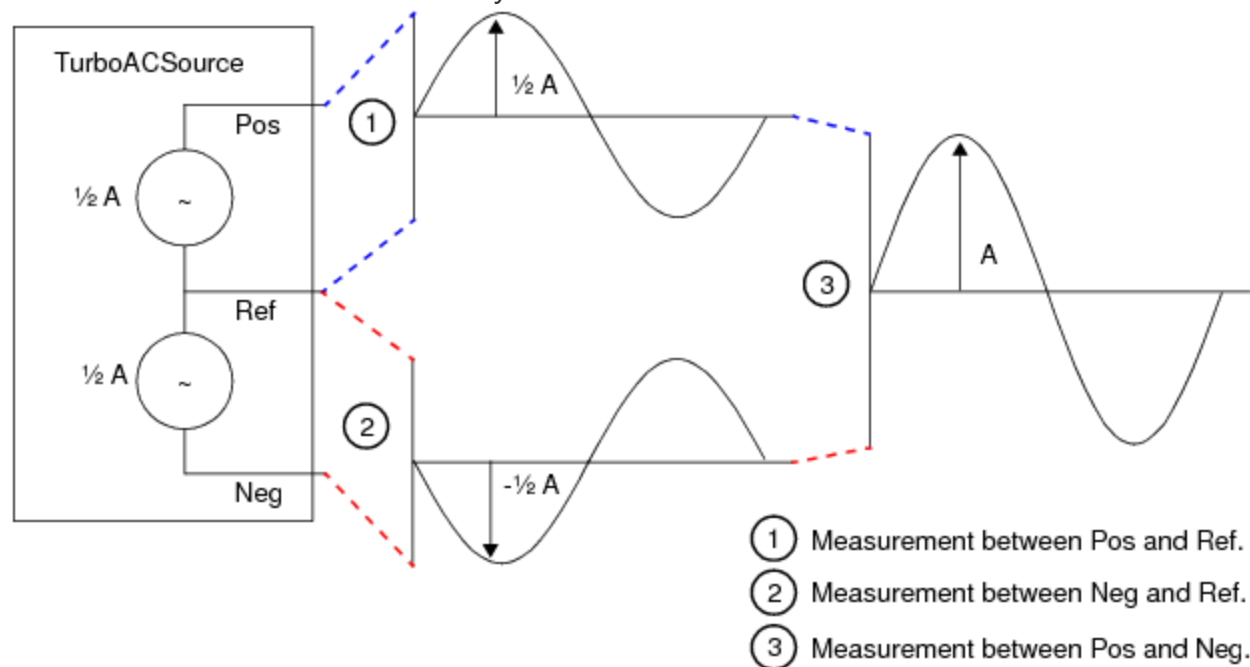
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TurboACSource Theory of Operation : [TurboACSource Overview](#) : Differential and Single-Ended Measurements

Differential and Single-Ended Measurements

The figure shows the TurboACSource instrument where A is the programmed amplitude. The amplitude represents the differential voltage, measured between the positive (Pos) and negative (Neg) channels. We say, therefore, that the peak voltage (V_{pk}) is A when measured single-ended and A when measured differentially.



Source Range Measurements

Likewise, the peak-to-peak voltage (V_{pp}) is defined with respect to the reference as the maximum voltage minus the minimum voltage over time signal. It is $1A$ when measured single-ended with respect to the reference and $2A$ when measured differentially between the positive and negative pins.

Since the TurboACSource can be either a differential or singled-ended instrument depending on the connection type, TurboACSource ranges and amplitudes are automatically scaled depending on the specified connection type.

TurboACSource ranges are always expressed in differential voltage. The amplitude depends on the connection type.

The source range specifies the maximum or minimum voltage that will be generated between the positive and negative channels, which is equivalent to a measurement in Vpk, differential. For example, if the amplitude is set to 1.28V, the range must be set to a value greater than or equal to 1.28V. Setting the range to a value smaller than the amplitude is invalid.

☒ **Note:**

AWG amplitude is programmed in Vpk into open circuit and CW amplitude is into the specified CW load.

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[TurboACSource Theory of Operation](#) : [TurboACSource Overview](#) : CC Common

CC Common

The CC Common is the channel card ground. CC Common can either be connected to the DUT or to the source reference.

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[TurboACSource Theory of Operation](#) : [TurboACSource Overview](#) : Device Ground Sense

Device Ground Sense

Device ground sense (DGS) is a sense signal that tracks the voltage of ground on the DUT and is used as a correction by the instrument to accurately force or measure voltages relative to the ground potential of the device. DGS must be connected to a single point on the DIB. The optimum place to connect DGS is at the device ground pin.

DGS connections are made by IG-XL when the instrument is connected to the DUT. Any used DGS must be grounded on the DIB or a DGS alarm can occur.

Note:

DGS pins are intended as a ground reference for the PPMUs and not for the TurboACSource instrument. Misapplication of a DGS connection can result in the instrument not realizing its performance specifications.

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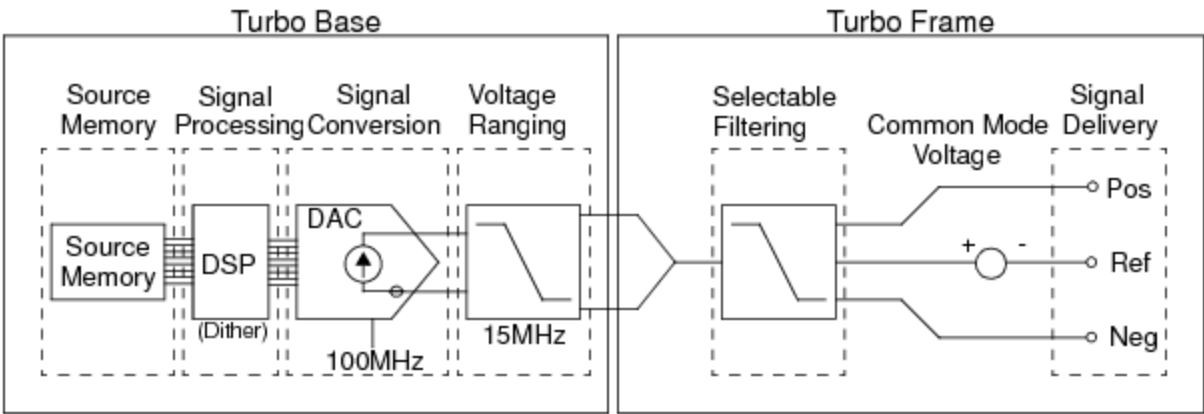
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TurboACSource Theory of Operation : AWG Source

AWG Source

The AWG Source instrument provides analog signals of a specific shape, frequency, and amplitude to a DUT in the frequency range of DC to 15MHz. This figure shows how the AWG Source hardware generates and delivers an analog signal.



Conceptual Diagram of the AWG Source

This process is described in the following topics:

- Source Memory
- Signal Processing
- Signal Conversion
- 15MHz Low-Pass Smoothing Filter
- Voltage Ranging

- [Dither](#)
- [Selectable Filtering](#)
- [External Trigger](#)

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TurboACSource Theory of Operation : [AWG Source](#) : Source Memory

Source Memory

The AWG Source memory (SMEM) is 4M samples deep. However, the largest single segment length is 2M samples. IG-XL loads source segments as required by host computer and pattern starts. At any one time, up to 500 segments can be loaded in SMEM.

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TurboACSource Theory of Operation : AWG Source : Signal Processing

Signal Processing

The AWG Source uses a series of digital signal processing stages to prepare the data for conversion by the DACs. First, the data is resampled to convert the sample rate from the user-specified sample rate to the instrument's fixed sample rate. The minimum programmable sample rate is 5ksps, and the maximum sample rate is 1Gsps. To achieve the instrument's performance specifications, however, the following rules determine the source sample rate and the number of samples in a source segment:

The minimum F_s allowed is:

$$(1) F_s \geq \text{Maximum [5 kHz or } 20 \times F_i]$$

Where F_i is the highest frequency of interest to be sourced.

The minimum N allowed is:

$$(2) N \geq \text{Maximum [20 or ceiling } \left(\frac{F_s}{100 \text{ Msp/s}} \right) \times 8]$$

Note:

The Mixed Signal Workshop (MSW) tool provides timing solutions that meet the instrument's sample rate and sample size requirements.

An amplitude scaling stage then adjusts the amplitude of the data. Calibration and noise-shaping are also applied. Finally, dither can optionally be added.

Adding dither to the source signal improves the signal distortion (THD and SFDR). Addition of dither tends to cause a slight increase in the noise density of the source. Since the dither is noise shaped, the noise density rises more at higher frequencies.

The default and reset state for dither is disabled.

	
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[TurboACSource Theory of Operation](#) : [AWG Source](#) : Signal Conversion

Signal Conversion

The [AWG Source](#) instrument’s digital-to-analog converter (DAC) converts digital samples to an analog signal at a fixed sample rate of 100 Msps.

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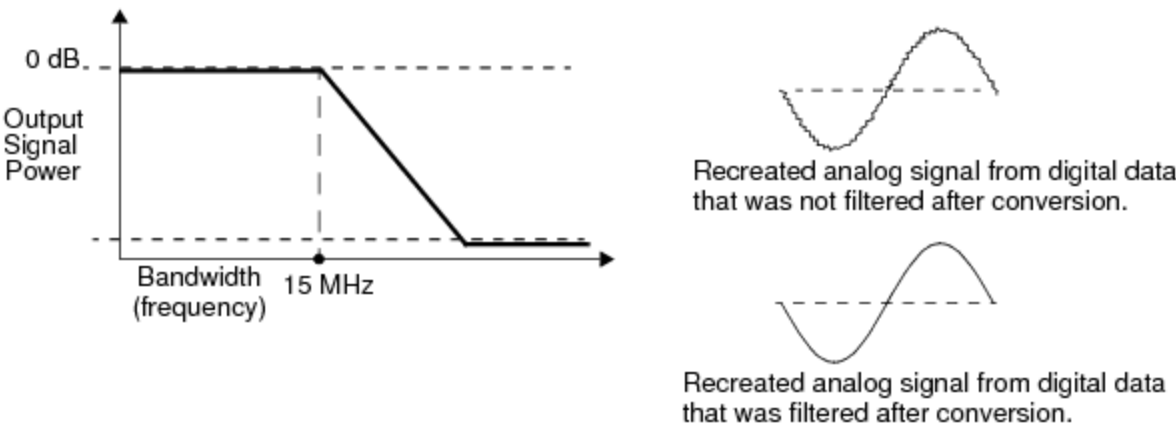
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TurboACSource Theory of Operation : AWG Source : 15MHz Low-Pass Smoothing Filter

15MHz Low-Pass Smoothing Filter

The source signal passes through a 15 MHz low-pass filter, which removes high frequency quantization noise from the synthesized signal and smooths the sequence of discrete analog levels into a continuous analog signal. Since the DAC always operates at 100 Msps, only one preset low-pass filter is required with pass band ending at 15 MHz (see 2-6). This filter is fixed and cannot be switched or controlled. With the conversion always running at a high speed, the noise shaping process does not generate large amounts of near out-of-band noise, thus producing a high fidelity signal.



15 MHz Low-Pass Filter Reduces High Frequency Quantization Noise

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TurboACSource Theory of Operation : [AWG Source](#) : Voltage Ranging

Voltage Ranging

The analog signal is attenuated by the range circuitry to achieve the programmed amplitude. The range circuitry is implemented as a resistor structure, which is designed to attenuate the full-scale voltage by -3dB increments, while maintaining a 50ohm output impedance.

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[TurboACSource Theory of Operation](#) : [AWG Source](#) : [Dither](#)

Dither

A TurboACSource instrument can add dither to the source signal to improve signal distortion (THD and SFDR). Addition of dither to the source signal slightly increases the noise density of the source. Since the dither is noise shaped, the noise density rises more at higher frequencies.

You enable and disable dither using VBT syntax. During debug, you can enable and disable dither using the TurboACSource debug display.

The following topics describe how to apply dither to a signal and how to control the dither hardware:

- [Enabling and Disabling Dither on a Signal](#)
- [Controlling Dither Hardware Directly](#)

Enabling and Disabling Dither on a Signal

The [TurboACSource Signals VBT property](#) .DitherEnabled allows you to enable or disable dither on a TurboACSource. Addition of dither to the source signal tends to slightly increase the noise density of the source. Since the dither is noise shaped, the noise density rises more at higher frequencies. The .DitherEnabled property defaults and resets to false.

Use the following syntax to enable dither for an AWG signal:

```
TheHdw.TurboACSource.Pins().AWG.Signals.Item().DitherEnabled.Value = True
```

Use the following syntax to disable dither for an AWG signal:

```
TheHdw.TurboACSource.Pins().AWG.Signals.Item().DitherEnabled.Value = False
```

Note:

[Setting](#) DitherEnabled.Value sets DitherEnabled.Mode to tlSignalModeUseValue.

[Controlling Dither Hardware Directly](#)

The TurboACSource Dither.Enabled property of the Pins interface allows you to directly enable or disable the dither hardware. Addition of dither to the source signal tends to slightly increase the noise density of the source. Since the dither is noise shaped, the noise density rises more at higher frequencies. The DitherEnabled property defaults and resets to false.

Use the following syntax to enable dither for the AWG:

TheHdw.TurboACSource.Pins().AWG.DitherEnabled = True

Use the following syntax to disable dither for the AWG:

TheHdw.TurboACSource.Pins().AWG.DitherEnabled = False

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TurboACSource Theory of Operation : [AWG Source](#) : Selectable Filtering

Selectable Filtering

The filters are conceptually divided in two categories:

- | | |
|---|---------------------|
| • | Buffered outputs |
| • | Nonbuffered outputs |

For filter settings under 25MHz, a buffer stage isolates the filters from the DUT load, as well as summing in the baseline offset voltage and ACDGS. Due to the performance requirements of the higher frequency filters ($\geq 25\text{MHz}$), the filter outputs are not buffered. When using these filters the DUT load must be 50ohms single-ended or 100ohms differential. If not, then termination must be used at the DIB.

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TurboACSource Theory of Operation : [AWG Source](#) : External Trigger

External Trigger

The TurboACSource board includes an external trigger pin for each TurboAC AWG source, for a total of four pins per board. The external trigger pin is an input pin that allows you to start sourcing a signal based on an external event. You can use the external trigger functionality to synchronize the TurboAC AWG source with other instruments in the tester by sending a signal from the DIB to any of the two AWG sources on the TurboAC.

Only use an external trigger when you cannot use pattern control for instrument/DUT synchronization. When using an external trigger, the start or trigger is synchronized to an external rising edge of a pulse from a digital channel that is presented to the External Trigger input of a TurboAC AWG.

You control the external trigger pins using VBT language statements that allow you to select microcodes (like START, TRIG) to be executed when an event occurs at the trigger input.

```
TheHdw.TurboACSource.Pins(PinList).AWG.ExternalTrigger.Load(tlTurboACSourceCommandStart, "MySignal")
```

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[TurboACSource Theory of Operation](#) : Standard CW Source

Standard CW Source

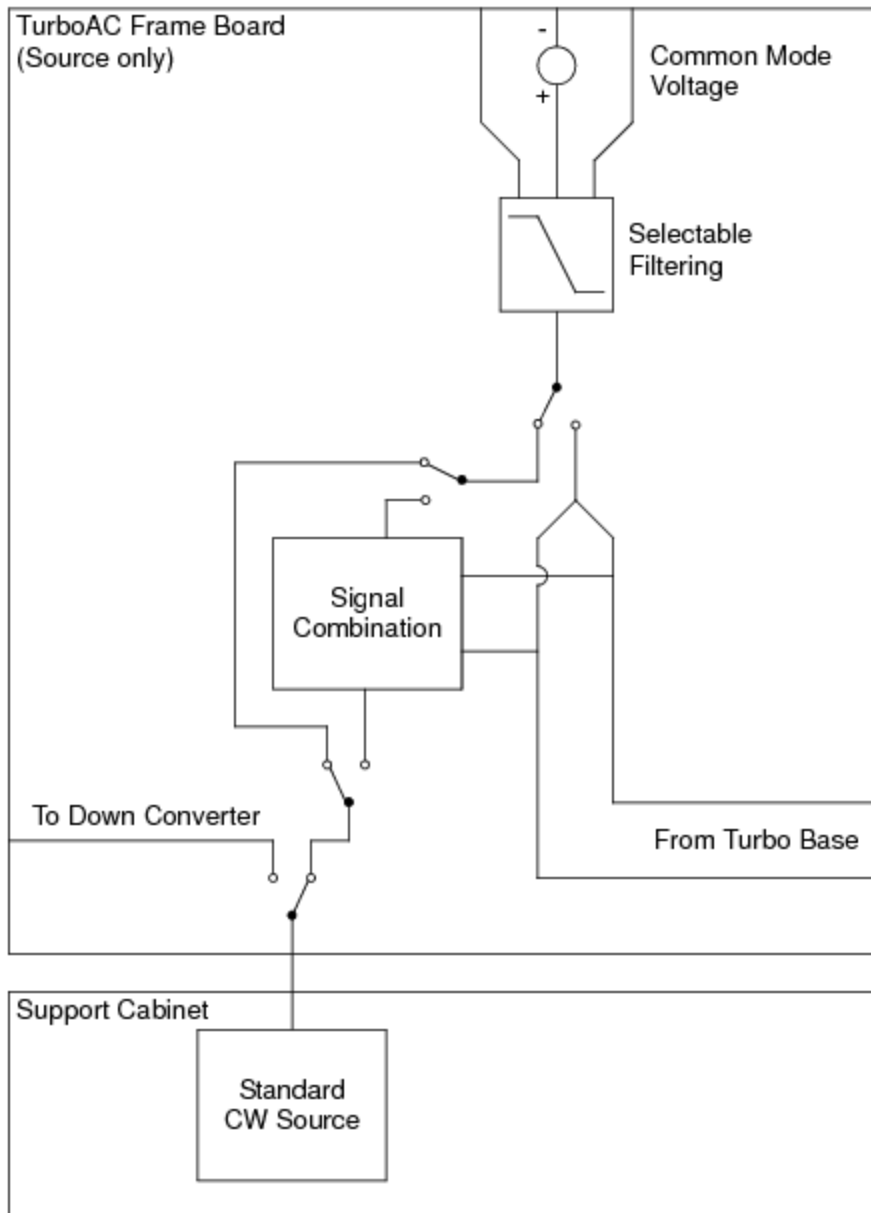
The Standard CW signal is generated by a synthesizer housed in the mainframe support cabinet. The signal travels from the support cabinet to the TurboAC Frame board through coaxial cable. The source is used to create CW signals for the source instrument in Standard CW mode and is also used to create a LO signal for the TurboACCapture's down conversion circuitry.

The TurboACSource in CW mode has a frequency range of 1MHz to 150MHz. The CW synthesizer interfaces to the test head through a 50ohm coaxial cable. The Synthesizer interface module provides a programming interface between the test system and the synthesizer.

Program the Standard CW source using the following VBT syntax:

```
TheHdw.TurboACSource.Pins(PinList).CW
```

The figure shows the connections between the Standard CW and the TurboAC Frame board.



Standard CW Source Block Diagram

For more information, see [Making a Differential Connection to Capture Cancellation Circuitry](#).

☒ **Note:**

The CW synthesizer is a shared resource that can be switched between the CW channel and the down converter LO. Addition of an optional synthesizer enables simultaneous operation of the down converter and CW channels on a single instrument.



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TurboACSource Theory of Operation : Optional CW Source

Optional CW Source

The TurboAC hardware set can accommodate an additional user-supplied CW synthesizer. When programmed in Optional CW source mode, the TurboACSource can use the additional synthesizer to create and apply single-tone sinusoidal analog signals to the DUT.

The following procedure outlines how to program the TurboACSource using a user supplied optional CW source:

1. Set up the instrument channels and connections.
- a. Map the TurboACSource channels to DUT pins in the pin map.

b. Connect the TurboACSource channels to resources or the DUT.

c. Connect any DUT pins to ground.

d. Set the signal's input source to optional CW.
2. Set the DC Offset.

Since the auxiliary input configuration closely resembles CW mode, use the AC parameter syntax for CW mode to program the common mode DC offset in Optional mode.

3. Specify the output mode as single-ended or differential.

Use the output parameter to specify whether to deliver the signal as a differential or single-ended output to the DUT.

4. Set the filters.

The filter settings available in AUX mode are: 1.2M, 6.1M, 12.1M, 25M, 48M, 70M, and 150M

5. Program the amplitude of the auxiliary synthesizer.

Using GPIB commands or some other means, program the amplitude of the auxiliary synthesizer.

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TurboACSource Theory of Operation : DIB Source

DIB Source

The TurboACSource’s DIB source mode allows signals from other test system instruments or external resources routed through the DIB to use the TurboACSource’s filter bank and signal combination slot. The DIB source mode input is a differential pair, and it is 50-ohm terminated on each line. For connecting a single-ended instrument to the DIB input, ground the unused input on the DIB.

The following procedure outlines how to program the TurboACSource in DIB mode:

1. Set up the instrument channels and connections.
- a. Map the TurboACSource channels to DUT pins in the pin map.

b. Connect the TurboACSource channels to resources or the DUT.

c. Connect any DUT pins to ground.

d. Set the signal’s input source to DIB.
2. Set the DC Offset.

Since the DIB input configuration closely resembles CW mode, use the AC parameter syntax for CW mode to program the common mode DC offset in AUX mode.

3. Specify output setting as single-ended or differential.

Use the output parameter to specify whether to deliver the signal as a differential or single-ended output to the DUT.

4. Set the filters.

The filter settings available in DIB mode are: 1.2M, 6.1 M, 12.1M, 25 M, 48 M, 70M, and 150 M.

5. Program the amplitude and start the instrument behind the DIB connection.

The instrument behind the DIB input connection to the TurboACSource will have its own procedure for setting the signal’s amplitude and starting the generation of the signal.

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TurboACSource Theory of Operation : Signal Combination

Signal Combination

The signal combination circuit allows you to combine two signals from any of the signal sources available to the TurboACSource instrument:

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|---|----------------|
| • | AWG + StdCW |
| • | AWG + DIBSrc |
| • | AWG + OptCW |
| • | StdCW + DIBSrc |
| • | StdCW + OptCW |
| • | OptCW + DIBSrc |

Signal combination is intended primarily for creating a dual tone signal, although it is not restricted to this purpose. Dual tones are often used for intermodulation distortion (IMD) testing. Creating a dual tone from two independent sources reduces the amount of IMD in the sourced signal so that any measured IMD comes chiefly from the DUT.

The signal combination circuit uses a 0 degree power combiner to create the combination signal. For optimum performance, enable the buffer and filters in the source signal path. When the output of the combination circuit is directly connected to the DUT, the output is a single-ended or differential signal that must be terminated in 50ohms or 100ohms for differential on the DIB.

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